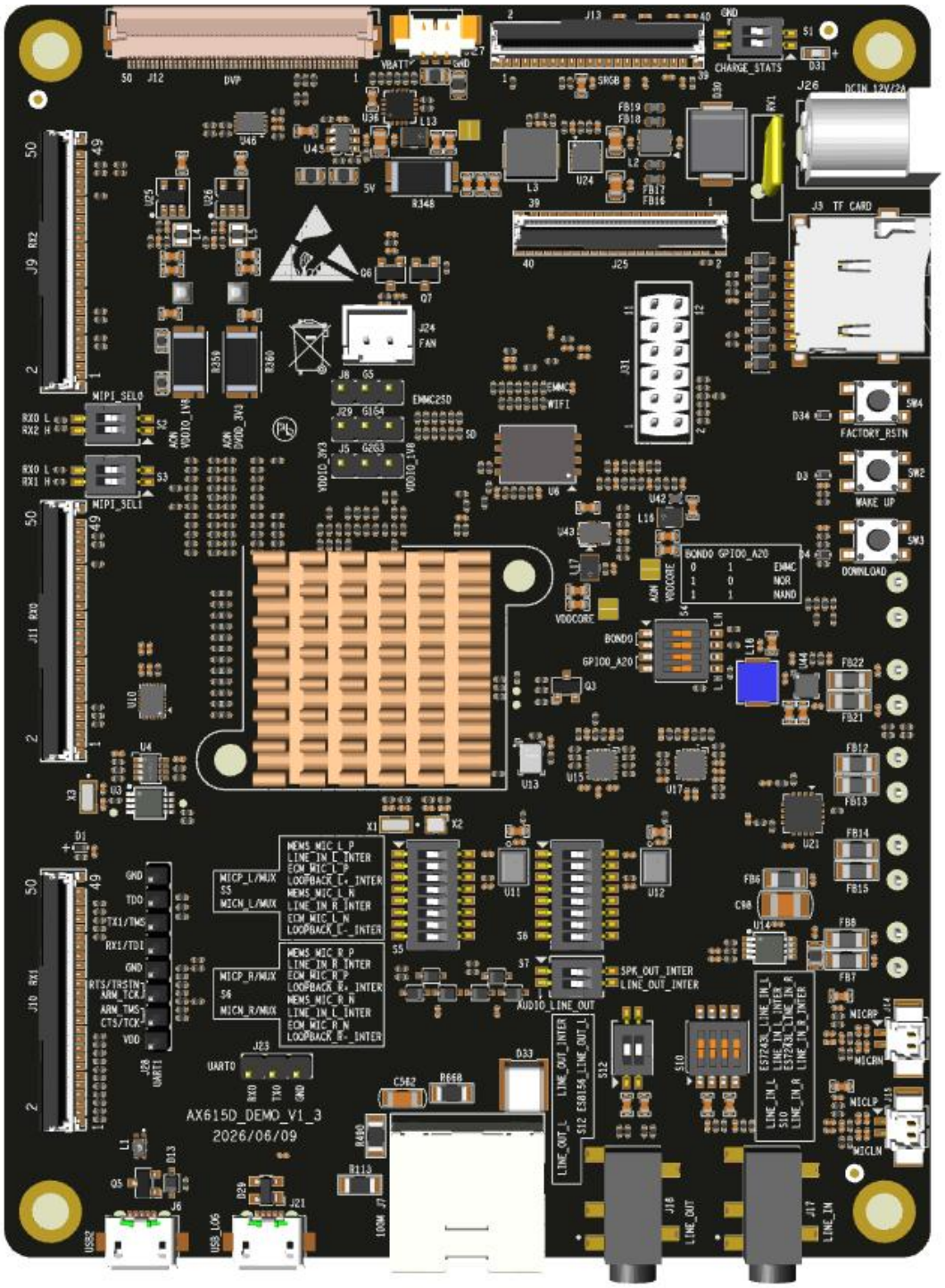
AX615D DEMO
VER 1.3

01. Revision History
02. System Diagram
03. Power Tree
04. I2C Map
06. Decap
05. SYS
07. SD CARD
08. Flash
09. ETH_USB
10. MIPI_IO
11. CAM SENSOR
12. DVP_BT_IN
13. SRGB_BT_OUT_LENS
14. AUDIO
15. EXT CODEC
16. LINE_MIC_IN_SPEAKER_OUT
17. DEBUG
18. POWER IN
19. POWER AON IO
20. POWER AON CORE
21. POWER MAIN IO
22. POWER MAIN CORE
23. BAT CHARGER



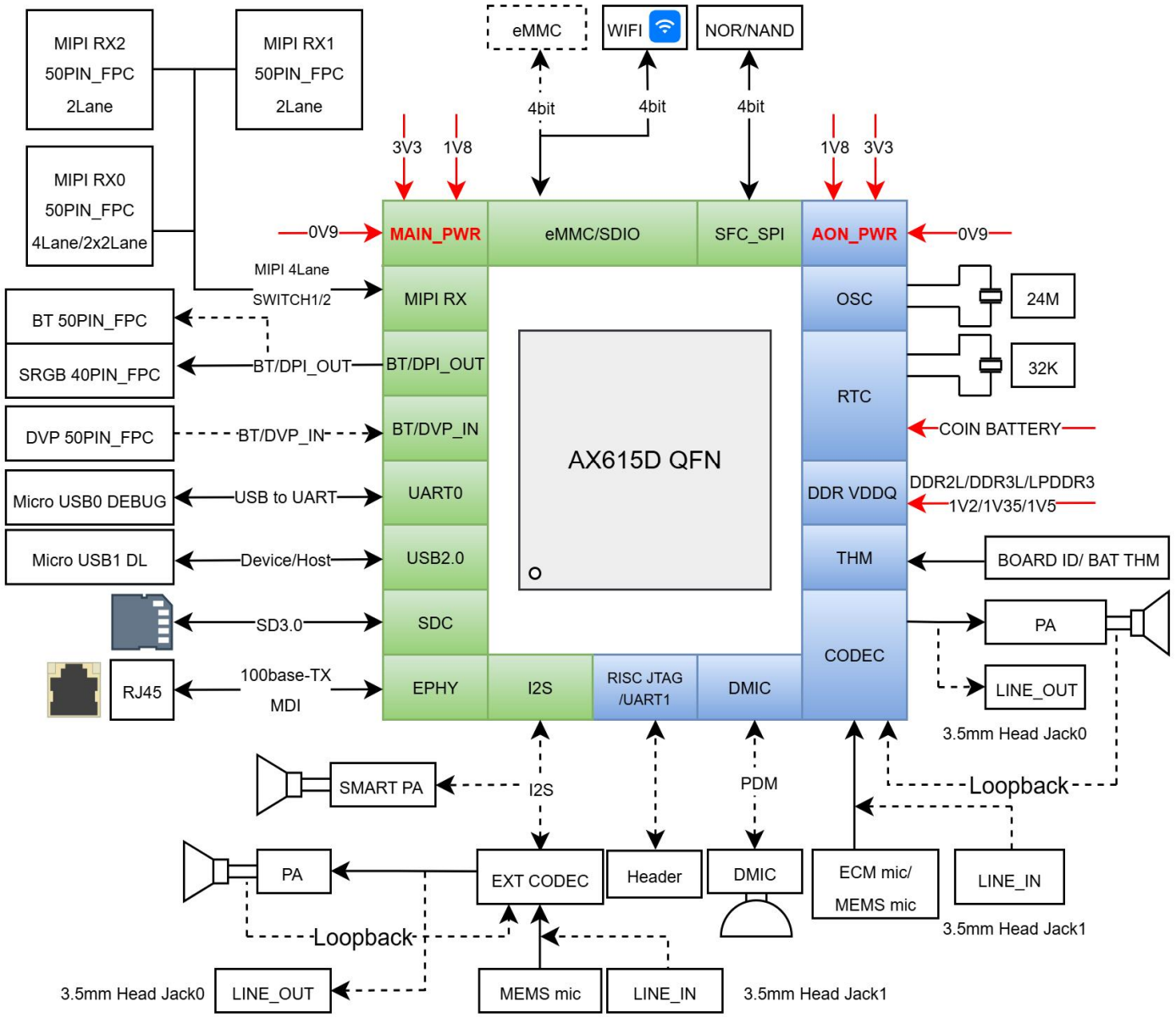
Revision History

Version	Date	Author	Description
V1.0	2025-05-20	xie yibei	Create Project
V1.1	2025-08-14	xie yibei	1.PAGE5 SOC power supply decap NC. 2.PAGE8 default used NAND Flash and WP pull up 4.7k. 3.PAGE7&PAGE8 add SD CARD & SDIO WIFI Jump circuit. 3.PAGE7&PAGE17 add MAIN ARM JTAG jump circuit. 4.PAGE9 replace ETH power supply for power monitor. 5.PAGE11 RX0 add AON I2C0 Jump. 6.PAGE13 BT656 OUT add I2C2 signal. 7.PAGE21 NC R662 for MAIN power down. 8.PAGE23 charger IC add OVP circuit.
V1.1.1	2025-09-19	xie yibei	1.PAGE9 EPHY signal add TVS and replace D33 part number for EOS/ESD. 2.PAGE10 R748 SMT,R126 NC,default used VDDCORE_DVS 3.PAGE19 J33 remove,add P20 NM_PASTE,use external DCDC to supply power to AON_DDRV_1V35 4.PAGE2&3 update power tree and system diagram about DDRVDDQ_1V35 and VDDCORE PN.
V1.1.2	2025-11-19	xie yibei	1.PAGE02&03&19 add DDR VDDQ note for DDR2 and DDR3L.
V1.2	2025-12-10	xie yibei	1.PAGE6 R755 SMT,R712 NC,default used AON_VDDCORE_DVS 2.PAGE7 change SDCARD IO use the EMMC group,and SDCARD power supply from AON_DVDD_3V3. 3.PAGE18 change DC JACK PN.
V1.2.1	2026-01-05	xie yibei	1.PAGE02&03&19 add DDR VDDQ note for DDR2\ DDR3L\LPDDR3.
V1.3	2026-06-09	xie yibei	1.PAGE19 add DDR VDDQ note for LPDDR2/DDR3, default LPDDR2. 2.PAGE06&11 add VSync and PWDN circuits for CAM Sensor . 3.PAGE18 change Nor Flash default. 4.PAGE23 NC charger circuits IC. 5.PAGE3 add Power Sequence.

AXERA

Page Name: Revision Histroy

Size	Project Name - Rev	Designer: xieyibei@axera-tech.com
A3	AX615D_DEMO	Checker: zhaopengfei@axera-tech.com
Date: Friday, June 12, 2026		Sheet 1 of 23

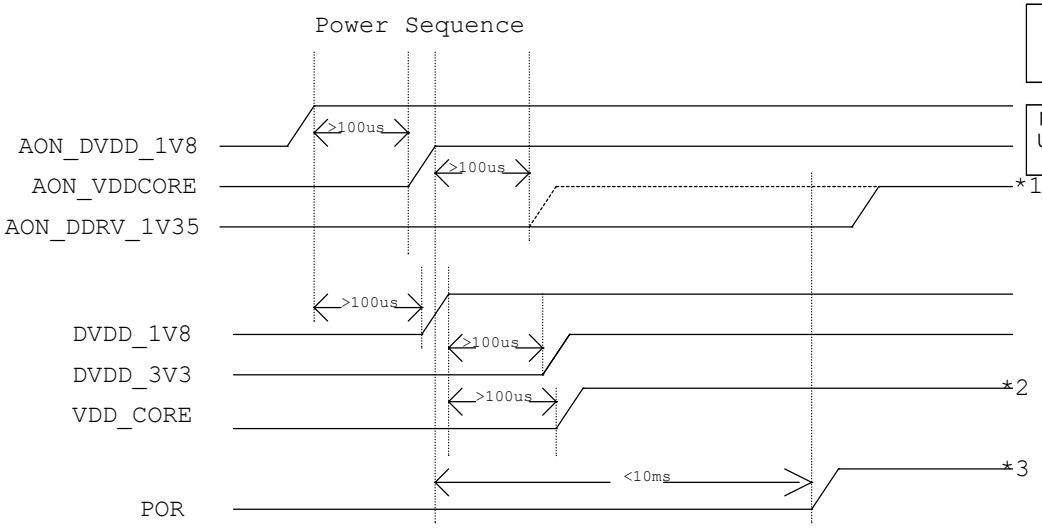


AXERA

Page Name: System Diagram

Size	Project Name - Rev	Designer: xieyibei@axera-tech.com
A3	AX615D_DEMO	Checker: zhaopengfei@axera-tech.com
Date: Tuesday, June 09, 2026		Sheet 2 of 23

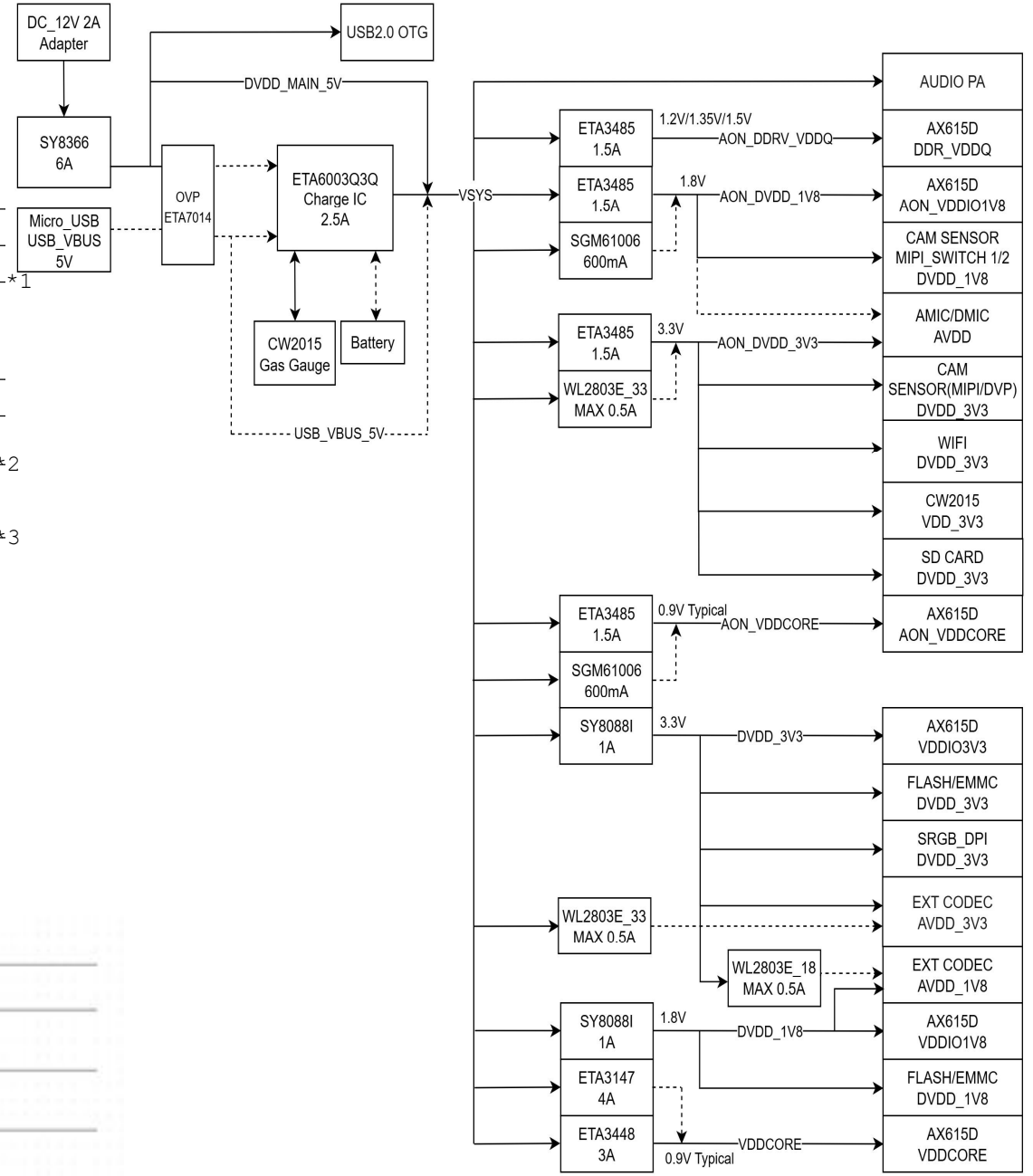
AOV方案时序



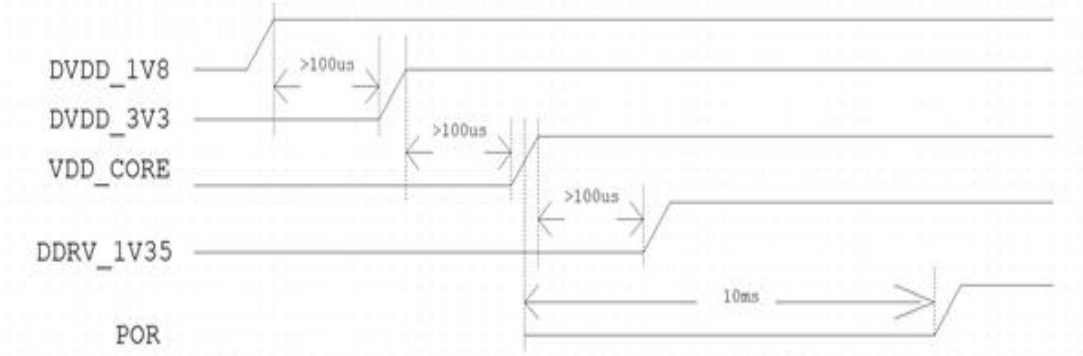
*1:外置AON_DDRV_1V35上电须晚于AON_VDDCORE

*2:VDD_CORE需晚于AON_VDDCORE上电

*3:除AON_DDRV_1V35外其他电源须在POR前完成上电



长电方案时序



POR在检测到VDDCORE达到阈值开始计时10ms后输出片内复位信号

AXERA

Page Name: Power Tree

Size	Project Name - Rev	Designer	Checker
A3	AX615D_DEMO	xieybei@axera-tech.com	zhupengfei@axera-tech.com

Date: Friday, June 12, 2026 Sheet 3 of 23

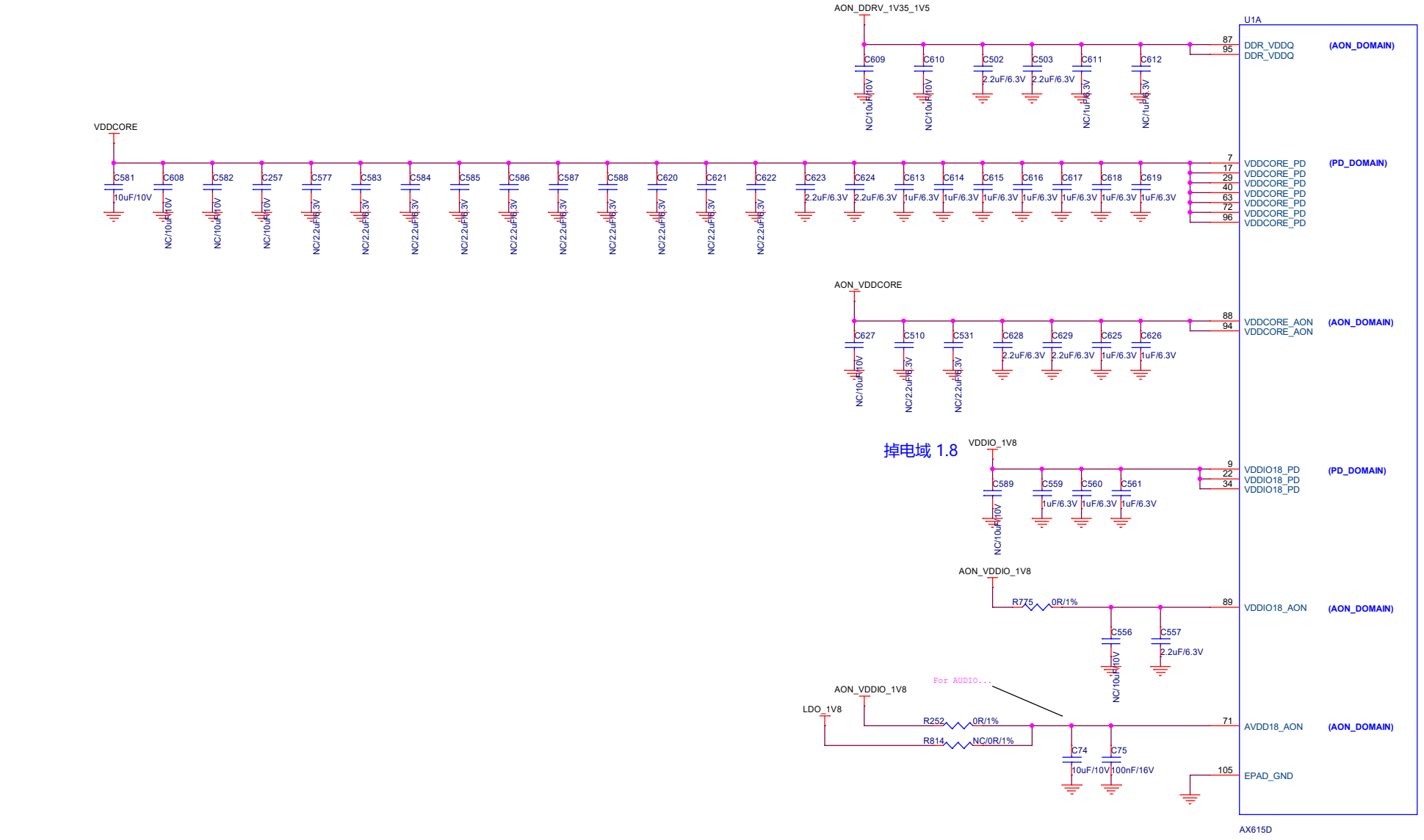
I2C MAP

I2C BUS	Host Domain	Device Domain	Device	Device Addr		Remark
				WRITE	READ	
AON_I2C0	1.8V	1.8V	CW2015	0xC5	0xC4	Gas Gauge
		1.8V	TBD	TBD	TBD	Camera Sensor1
AON_I2C1	1.8V	1.8V	TBD	TBD	TBD	Camera Sensor0/2
		1.8V	BT IN/OUT	TBD	TBD	DVP_BT IN
I2C2	1.8V/3.3V	1.8V	ES7243L	0x2D	0x2C	EXT ADC
		1.8V	ES8156	0x13	0x12	EXT DAC
		3.3V	AT8563S	0xA3	0xA2	RTC chip
		1.8V	AW88261	0x6B	0x6A	SMART PA
		3.3V	TBD	TBD	TBD	BT656 OUT

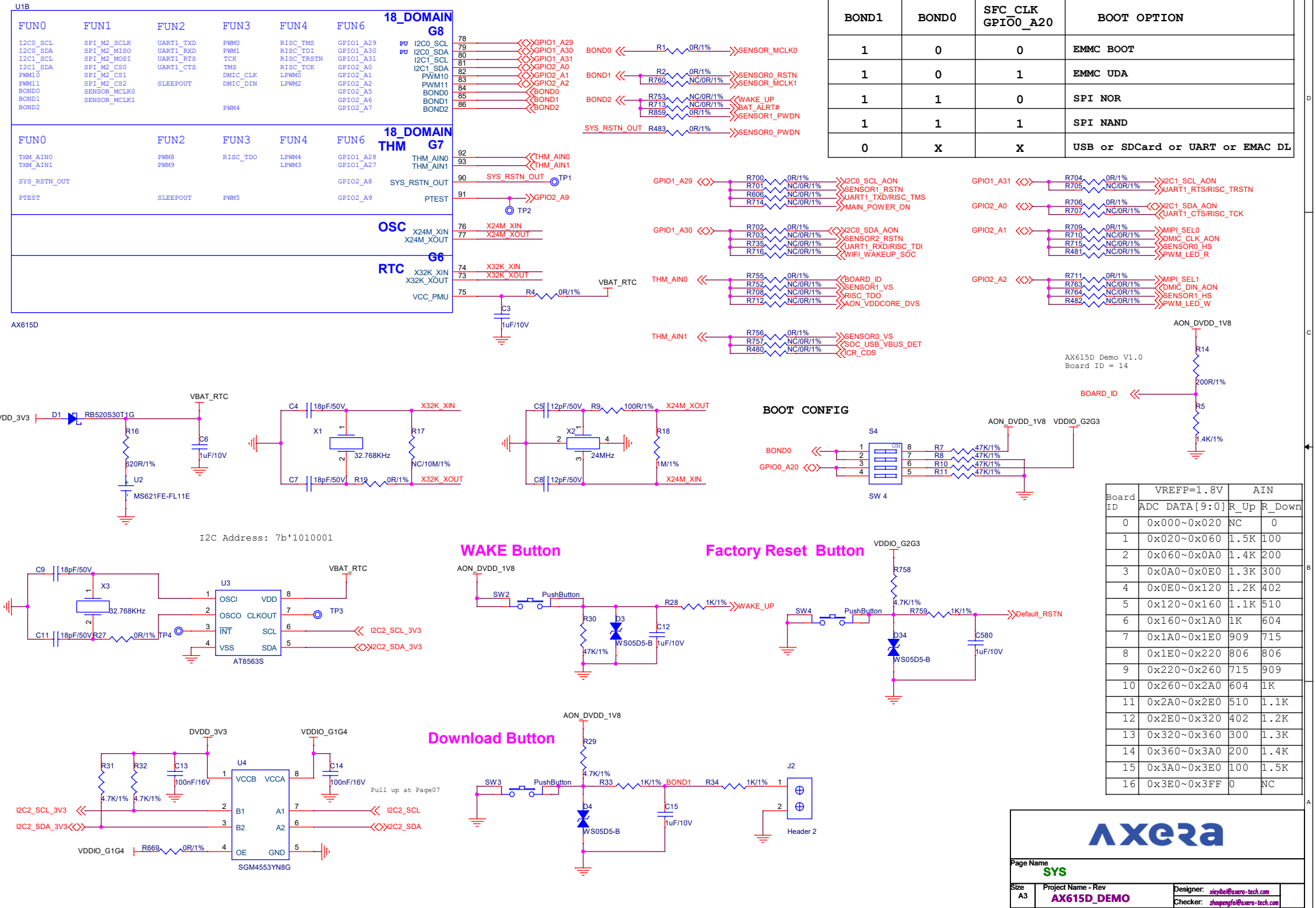
AXERA

Page Name
I2C MAP

Size	Project Name - Rev	Designer: xieyibei@axera-tech.com
A4	AX615D_DEMO	Checker: zhoupengfei@axera-tech.com
Date: Tuesday, June 09, 2026		Sheet 4 of 23



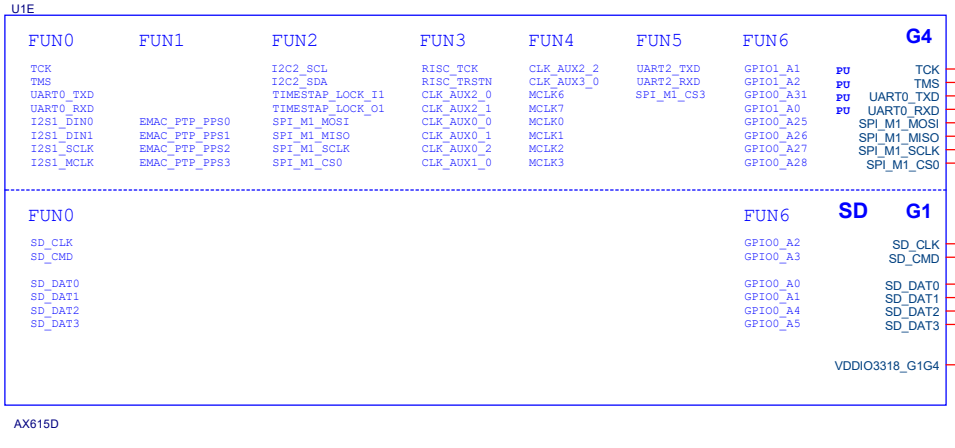
Page Name: Decap			
Size: A3	Project Name - Rev: AX615D_DEMO	Designer: xieyibo@axera-tech.com	
Date: Tuesday, June 09, 2026		Checker: zhoupengfei@axera-tech.com	Sheet 5 of 23

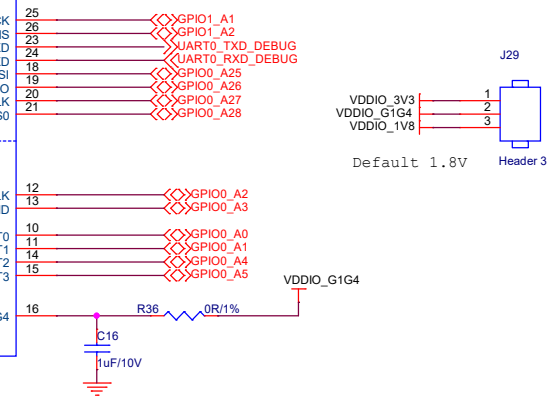


BOND1	BOND0	SFC_CLK GPIO0_A20	BOOT OPTION
1	0	0	EMMC BOOT
1	0	1	EMMC UDA
1	1	0	SPI NOR
1	1	1	SPI NAND
0	X	X	USB or SDCard or UART or EMAC DL

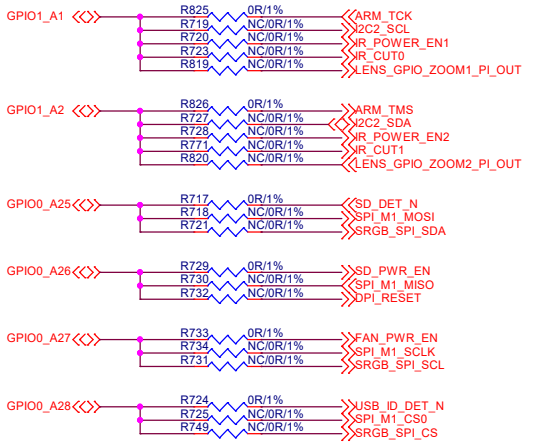
Board ID	VREFP=1.8V ADC DATA[9:0]	AIN R_Up	AIN R_Down
0	0x000~0x020	NC	0
1	0x020~0x060	1.5K	100
2	0x060~0x0A0	1.4K	200
3	0x0A0~0x0E0	1.3K	300
4	0x0E0~0x120	1.2K	402
5	0x120~0x160	1.1K	510
6	0x160~0x1A0	1K	604
7	0x1A0~0x1E0	909	715
8	0x1E0~0x220	806	806
9	0x220~0x260	715	909
10	0x260~0x2A0	604	1K
11	0x2A0~0x2E0	510	1.1K
12	0x2E0~0x320	402	1.2K
13	0x320~0x360	300	1.3K
14	0x360~0x3A0	200	1.4K
15	0x3A0~0x3E0	100	1.5K
16	0x3E0~0x3FF	0	NC

Page Name: SYS
Size: A3
Project Name - Rev: AX615D_DEMO
Designer: xieyibei@axera-tech.com
Checker: zhupengfei@axera-tech.com
Date: Tuesday, June 09, 2026
Sheet 6 of 23

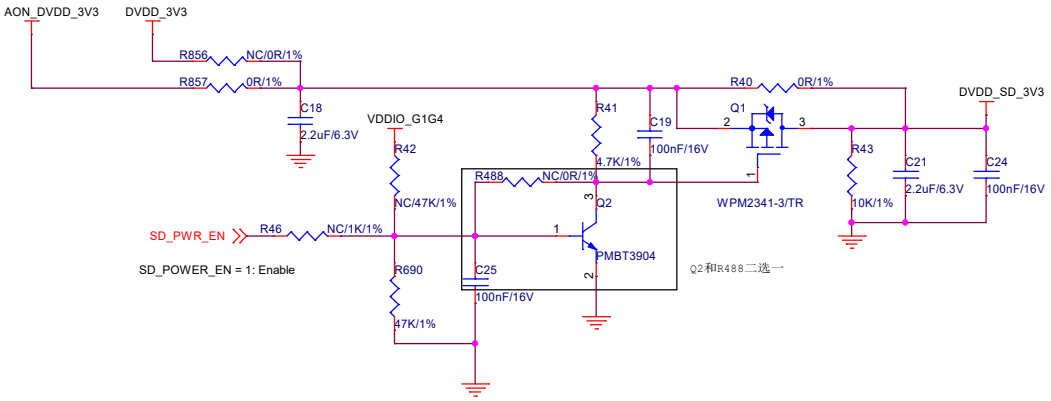


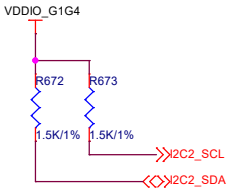


G4

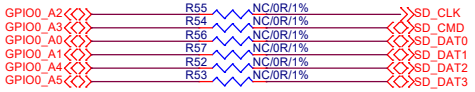


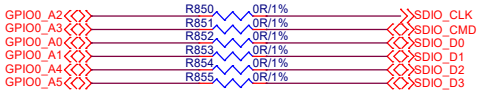
SD CARD: G2 EMMC group by default



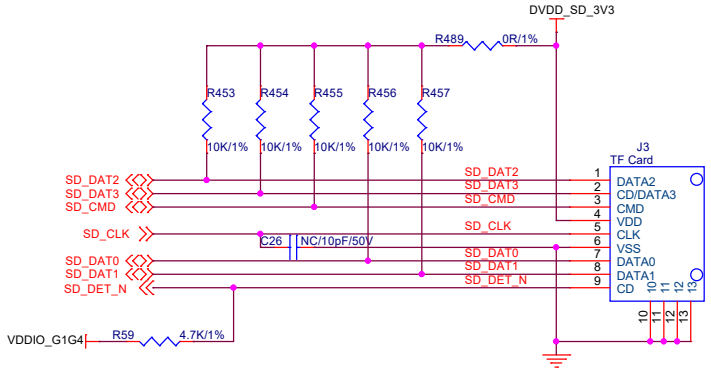


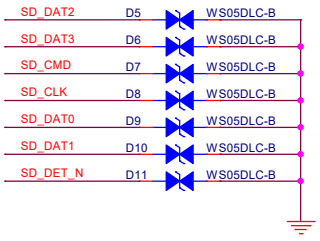
SD CARD & SDIO WIFI Jump





使用SD CARD功能请确保VDDIO_G1G4/G2G3的电源是3.3V





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Page Name: SD_CARD

Size	Project Name - Rev	Designer: xieyibei@axera-tech.com
A3	AX615D_DEMO	Checker: zhangengfei@axera-tech.com
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U1G

FUN0 / FUN6 / G3

SFC_CLK	GPIO0_A20	SFC_CLK	101 GPIO0_A20
SFC_CSN0	GPIO0_A18	SFC_CSN0	103 GPIO0_A18
SFC_CSN1	GPIO0_A19	SFC_CSN1	102 GPIO0_A19
SFC_MOSI_IO0	GPIO0_A21	SFC_MOSI_IO0	100 GPIO0_A21
SFC_MISO_IO1	GPIO0_A22	SFC_MISO_IO1	99 GPIO0_A22
SFC_WP_IO2	GPIO0_A23	SFC_WP_IO2	98 GPIO0_A23
SFC_HOLD_IO3	GPIO0_A24	SFC_HOLD_IO3	97 GPIO0_A24

FUN0 / FUN2 / FUN3 / FUN6 / G2

EMMC_CLK	I2S0_SCLK		GPIO0_A8	EMMC_CLK 3 GPIO0_A8
EMMC_CMD	I2S0_MCLK		GPIO0_A9	EMMC_CMD 2 GPIO0_A9
EMMC_DAT0	I2S0_DIN0		GPIO0_A6	EMMC_DAT0 104 GPIO0_A6
EMMC_DAT1	I2S0_DIN1		GPIO0_A7	EMMC_DAT1 1 GPIO0_A7
EMMC_DAT2	I2S0_LRCK		GPIO0_A10	EMMC_DAT2 5 GPIO0_A10
EMMC_DAT3	I2S0_DOUT		GPIO0_A11	EMMC_DAT3 6 GPIO0_A11
EMMC_RESET_N	I2S0_MCLK	WDT_THM_RST_N	GPIO0_A17	EMMC_RESET_N 8 GPIO0_A17

VDDIO3318_G2G3 4 — R73 0R/1% — VDDIO_G2G3; C29 1uF/10V

AX615D

GPIO0_A19 — R765 NC/0R/1% — Default_RSTN; R738 0R/1% — WIFI_RSTN

J5: VDDIO_3V3 1, VDDIO_G2G3 2, VDDIO_1V8 3 — Default 3.3V — Header 3

SPI FLASH

DVDD_FLASH

GPIO0_A20 R60 0R/1% SFC_CLK
GPIO0_A18 R62 0R/1% SFC_CSN0
GPIO0_A21 R63 0R/1% SFC_MOSI_IO0
GPIO0_A22 R66 0R/1% SFC_MISO_IO1
GPIO0_A23 R67 0R/1% SFC_WP_IO2
GPIO0_A24 R68 0R/1% SFC_HOLD_IO3

DVDD_1V8 R61 NC/0R/1%
DVDD_3V3 R64 0R/1%

DVDD_FLASH

R69 4.7K/1% R70 4.7K/1% R71 4.7K/1% C28 2.2uF/6.3V

U6: W25Q128JVEIQ

SFC_CSN0	1	CS	VCC 8
SFC_MISO_IO1	2	DO/IO1	HOLD/IO3 7 SFC_HOLD_IO3
SFC_WP_IO2	3	WP/IO2	CLK 6 SFC_CLK
	4	GND	DI/IO0 5 SFC_MOSI_IO0
	9	PowerPad	

R72 NC/4.7K/1%

EMMC

GPIO0_A8 R74 NC/0R/1% EMMC_CLK
GPIO0_A9 R76 NC/0R/1% EMMC_CMD
GPIO0_A6 R77 NC/0R/1% EMMC_D0
GPIO0_A7 R78 NC/0R/1% EMMC_D1
GPIO0_A10 R79 NC/0R/1% EMMC_D2
GPIO0_A11 R75 NC/0R/1% EMMC_D3
GPIO0_A17 R88 NC/0R/1% EMMC_RESET_N

DVDD_EMMC_VCCQ

DVDD_1V8 R95 0R/1%
DVDD_3V3 R96 NC/0R/1%

U7A: KLM8G1GETF-B041

EMMC_D0	A3	D0	VCCQ C6
EMMC_D1	A4	D1	VCCQ M4
EMMC_D2	A5	D2	VCCQ N4
EMMC_D3	B2	D3	VCCQ P3
	B3	D4	VCCQ P5
	B4	D5	
	B5	D6	VCC K9
	B6	D7	VCC J10
	H5	Data Strobe	VCC F5
EMMC_CLK	M6	CLK	VCC E6
EMMC_CMD	M5	CMD	VSSQ C4
EMMC_RESET_N	K5	RSTN	VSSQ N2
VDDI	C2	VDDI	VSSQ N5
			VSSQ P4
			VSSQ P6
			VSSQ E7
			VSS G5
			VSS H10
			VSS K8
			VSS A6
			VSS J5

DVDD_EMMC_VCCQ — C30 100nF/16V, C31 100nF/16V, C32 100nF/16V, C33 2.2uF/6.3V

DVDD_3V3 — C34 100nF/16V, C35 2.2uF/6.3V

DVDD_EMMC_VCCQ — R93 10K/1% EMMC_CMD; R94 10K/1% EMMC_RESET_N

VDDI C2 — C36 1uF/10V

U7B: KLM8G1GETF-B041

H1	NC_R1	NC_P14	G14
H2	NC_R2	NC_P13	G13
H3	NC_R3	NC_P12	G12
H12	NC_R12	NC_P11	G10
H13	NC_R13	NC_P10	G3
H14	NC_R14	NC_P3	G2
J1	NC_T1	NC_P2	G1
J2	NC_T2	NC_P1	F14
J3	NC_T3	NC_N14	F13
J12	NC_T12	NC_N13	F12
J13	NC_T13	NC_N12	F10
J14	NC_T14	NC_N10	F3
K1	NC_U1	NC_N3	F2
K2	NC_U2	NC_N2	F1
K3	NC_U3	NC_N1	E14
K6	NC_U6	NC_M14	E13
K7	NC_U7	NC_M13	E12
K10	NC_U10	NC_M12	E10
K12	NC_U12	NC_M10	E9
K13	NC_U13	NC_M9	E8
K14	NC_U14	NC_M8	E5
L1	NC_V1	NC_M5	E3
L2	NC_V2	NC_M3	E2
L3	NC_V3	NC_M2	E1
L12	NC_V12	NC_M1	D14
L13	NC_V13	NC_L14	D13
L14	NC_V14	NC_L13	D12
M1	NC_W1	NC_L12	D4
M2	NC_W2	NC_L4	D3
M3	NC_W3	NC_L3	D2
M7	NC_W7	NC_L2	D1
M8	NC_W8	NC_L1	C14
M9	NC_W9	NC_K14	C13
M10	NC_W10	NC_K13	C12
M11	NC_W11	NC_K12	C11
M12	NC_W12	NC_K11	C10
M13	NC_W13	NC_K10	C9
M14	NC_W14	NC_K9	C8
N1	NC_Y1	NC_K8	C7
N3	NC_Y3	NC_K7	C5
N6	NC_Y6	NC_K5	C3
N7	NC_Y7	NC_K3	C1 — VDDI
N8	NC_Y8	NC_K1	B14
N9	NC_Y9	NC_J14	B13
N10	NC_Y10	NC_J13	B12
N11	NC_Y11	NC_J12	B11
N12	NC_Y12	NC_J11	B10
N13	NC_Y13	NC_J10	B9
N14	NC_Y14	NC_J9	B8
P1	NC_AA1	NC_J8	B7
P2	NC_AA2	NC_J7	B1 — EMMC_D3
P7	NC_AA7	NC_J1	A14
P8	NC_AA8	NC_H14	A13
P9	NC_AA9	NC_H13	A12
P10	NC_AA10	NC_H12	A11
P11	NC_AA11	NC_H11	A10
P12	NC_AA12	NC_H10	A9
P13	NC_AA13	NC_H9	A8
P14	NC_AA14	NC_H8	A7
		NC_H7	A2
		NC_H2	A1
		NC_H1	

SD CARD & SDIO WIFI Jump

使用SDIO WIFI功能请确保VDDIO_G2G3的电源是3.3V，注意使用的FLASH电压为3.3V

GPIO0_A8 R83 NC/0R/1% SDIO_CLK
GPIO0_A9 R85 NC/0R/1% SDIO_CMD
GPIO0_A6 R87 NC/0R/1% SDIO_D0
GPIO0_A7 R89 NC/0R/1% SDIO_D1
GPIO0_A10 R90 NC/0R/1% SDIO_D2
GPIO0_A11 R91 NC/0R/1% SDIO_D3

GPIO0_A8 R844 0R/1% SD_CLK
GPIO0_A9 R845 0R/1% SD_CMD
GPIO0_A6 R846 0R/1% SD_DAT0
GPIO0_A7 R847 0R/1% SD_DAT1
GPIO0_A10 R848 0R/1% SD_DAT2
GPIO0_A11 R849 0R/1% SD_DAT3

WIFI_VDDIO_3V3

R458 R459 R460 R461 R462 10K/1%

J31: NC/Header2X6

SDIO_CLK	SDIO_CLK	1	12
SDIO_CMD	SDIO_CMD	2	11
SDIO_D0	SDIO_D0	3	10
SDIO_D1	SDIO_D1	4	9 — SOC_WAKEUP_WIFI
SDIO_D2	SDIO_D2	5	8 — WIFI_WAKEUP_SOC
SDIO_D3	SDIO_D3	6	7 — WIFI_RSTN

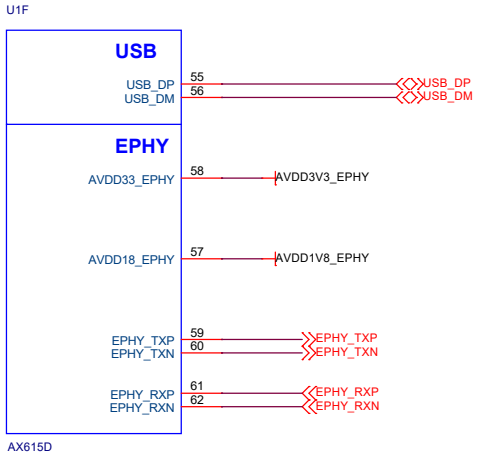
WIFI_DVDD_3V3

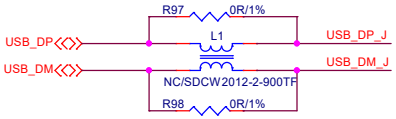
AON_DVDD_3V3 — R780 0R/1% — WIFI_DVDD_3V3; C415 10uF/10V, C440 100nF/16V

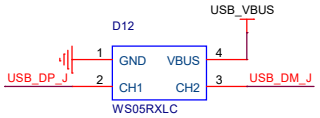
WIFI_VDDIO_3V3 — R824 NC/0R/1%; R463 NC/0R/1% — WIFI_DVDD_3V3 — DVDD_3V3

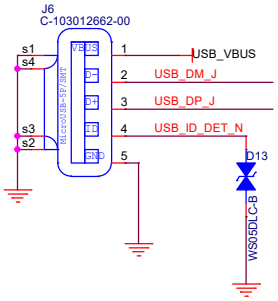
Page Name	Flash
Size	A3
Project Name - Rev	AX615D_DEMO
Designer:	xieyibei@axera-tech.com
Checker:	zhupengfei@axera-tech.com
Date:	Tuesday, June 09, 2026
Sheet	8 of 23

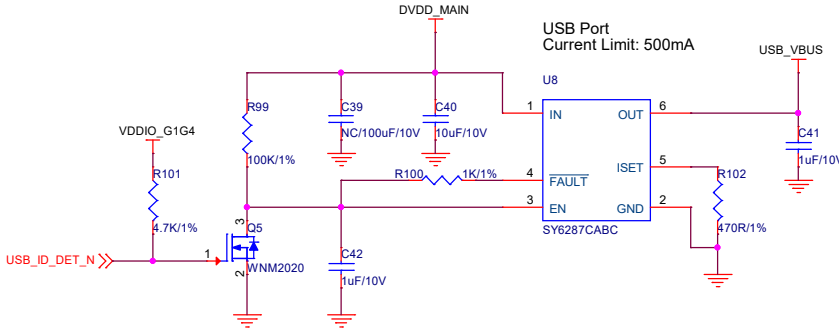
USB



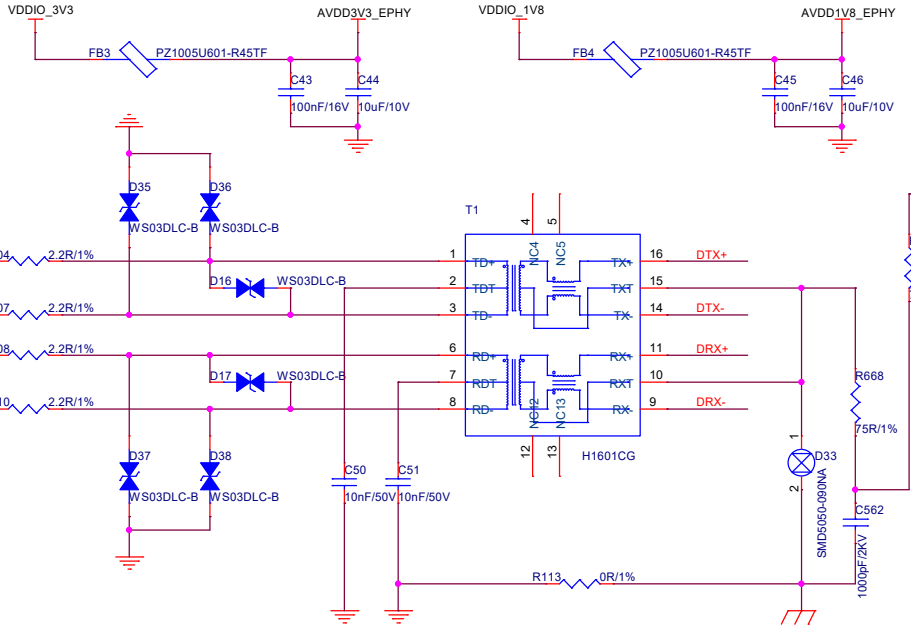




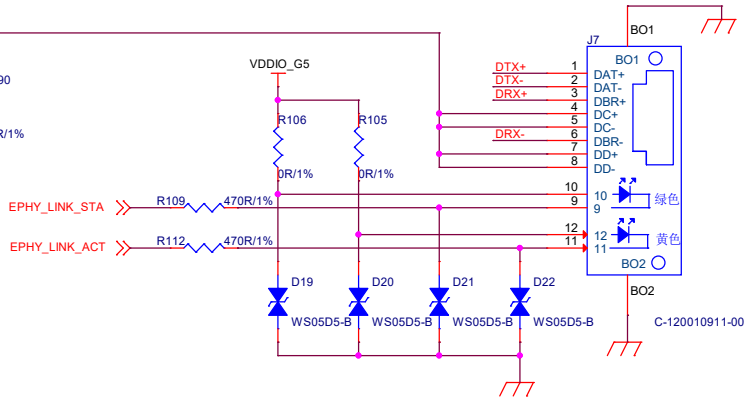




ETH



RJ45网络指示灯正常工作要求：VDDIO_G5需跳帽到3.3V



Page Name: ETH_USB			
Size: A3	Project Name - Rev: AX615D_DEMO	Designer: xieyibei@axera-tech.com	
		Checker: zhangengfei@axera-tech.com	
Date: Tuesday, June 09, 2026		Sheet 9 of 23	

U1D

FUN0	FUN1	FUN2	FUN4	FUN5	FUN6
VO_BT1120_CLK	VI_BT1120_D9		DMIC_CLK		GPIO1_A8
VO_BT1120_D0	VI_BT1120_D0	PWM6		I2S1_DIN0	GPIO1_A15
VO_BT1120_D1	VI_BT1120_D1	PWM7	DMIC_DIN	I2S1_DIN1	GPIO1_A16
VO_BT1120_D2	VI_BT1120_D2	UART0_RTS	LPWM6	I2S1_SCLK	GPIO1_A17
VO_BT1120_D3	VI_BT1120_D3	UART0_CTS	LPWM5	I2S1_MCLK	GPIO1_A18
VO_BT1120_D4	VI_BT1120_D4	SPI_M0_MOSI		TIMESTAMP_LOCK_O0	GPIO1_A19
VO_BT1120_D5	VI_BT1120_D5	SPI_M0_MISO		TIMESTAMP_LOCK_I0	GPIO1_A20
VO_BT1120_D6	VI_BT1120_D6	SPI_M0_SCLK			GPIO1_A21
VO_BT1120_D7	VI_BT1120_D7	SPI_M0_CS0			GPIO1_A22
VO_BT1120_D8	VI_BT1120_D8	PWM8	EPHY_LINK_ACT	I2S1_LRCK	GPIO1_A23
VO_BT1120_D9	VI_BT1120_CLK0	PWM9	EPHY_LINK_STA	I2S1_DOUT	GPIO1_A24

G5

GPIO1_A8 27 GPIO1_A8
PWM6 28 GPIO1_A15
PWM7 30 GPIO1_A16
UART0_RTS 31 GPIO1_A17
UART0_CTS 32 GPIO1_A18
SPI_M0_MOSI 35 GPIO1_A19
SPI_M0_MISO 36 GPIO1_A20
SPI_M0_SCLK 37 GPIO1_A21
SPI_M0_CS0 38 GPIO1_A22
EPHY_LINK_ACT 39 GPIO1_A23
EPHY_LINK_STA 41 GPIO1_A24

J8
VDDIO_3V3 1
VDDIO_G5 2
VDDIO_1V8 3
Default 1.8V Header 3

VDDIO3318_G5 33 R114 0R/1% VDDIO_G5
C53 1uF/10V

FUN0	FUN1
RX_CD0_P	VI_BT1120_D11
RX_CD0_N	VI_BT1120_D10
RX_CD1_P	VI_BT1120_D13
RX_CD1_N	VI_BT1120_D12
RX_CD2_P	VI_BT1120_D15
RX_CD2_N	VI_BT1120_D14
RX_CD3_P	VI_BT1120_D17
RX_CD3_N	VI_BT1120_D16
RX_CD4_P	VI_BT1120_D19
RX_CD4_N	VI_BT1120_D18
RX_CD5_P	VI_BT1120_D20
RX_CD5_N	VI_BT1120_CLK1

RX_CD0_P 42 RX_CD0_P
RX_CD0_N 43 RX_CD0_N
RX_CD1_P 44 RX_CD1_P
RX_CD1_N 45 RX_CD1_N
RX_CD2_P 46 RX_CD2_P
RX_CD2_N 47 RX_CD2_N
RX_CD3_P 49 RX_CD3_P
RX_CD3_N 50 RX_CD3_N
RX_CD4_P 51 RX_CD4_P
RX_CD4_N 52 RX_CD4_N
RX_CD5_P 53 RX_CD5_P
RX_CD5_N 54 RX_CD5_N

Diff 100Ω

VDDIO18_RX 48 AVDD18_RX R115 0R/1% VDDIO_1V8
C54 1uF/10V

AX615D

GPIO1_A8 R116 0R/1% VI_D9
R119 NC/0R/1% VO_CLK
R821 NC/0R/1% SOC_WAKEUP_WIFI

GPIO1_A15 R121 0R/1% VI_D0
R124 NC/0R/1% VO_D0
R739 NC/0R/1% I2S1_DIN0

GPIO1_A16 R748 0R/1% VDDCORE_DVS
R126 NC/0R/1% VI_D1
R129 NC/0R/1% VO_D1
R740 NC/0R/1% I2S1_DIN1

GPIO1_A17 R131 0R/1% VI_D2
R134 NC/0R/1% VO_D2
R741 NC/0R/1% I2S1_SCLK

GPIO1_A18 R137 0R/1% VI_D3
R139 NC/0R/1% VO_D3
R742 NC/0R/1% I2S1_MCLK

GPIO1_A19 R117 0R/1% VI_D4
R120 NC/0R/1% VO_D4
R747 NC/0R/1% LINE_IN_DET

GPIO1_A20 R122 0R/1% VI_D5
R125 NC/0R/1% VO_D5
R726 NC/0R/1% LINE_OUT_DET
R817 NC/0R/1% LENS_GPIO_PLSB

GPIO1_A21 R127 0R/1% VI_D6
R130 NC/0R/1% VO_D6
R722 NC/0R/1% SGM2820_EN
R818 NC/0R/1% LENS_GPIO_PLSA

GPIO1_A22 R132 0R/1% VI_D7
R135 NC/0R/1% VO_D7
R478 NC/0R/1% SMART_PA_INTN
R815 NC/0R/1% LENS_GPIO_VFA

GPIO1_A23 R138 0R/1% VI_D8
R140 NC/0R/1% VO_D8_HS
R743 NC/0R/1% I2S1_LRCLK
R745 NC/0R/1% EPHY_LINK_ACT

GPIO1_A24 R141 0R/1% VI_CLK0
R142 NC/0R/1% VO_D9_VS
R744 NC/0R/1% I2S1_DOUT
R746 NC/0R/1% EPHY_LINK_STA

RX_CD0_P R143 0R/1% MIPI_RX_CD0_P
R145 NC/0R/1% VI_D11

RX_CD0_N R147 0R/1% MIPI_RX_CD0_N
R149 NC/0R/1% VI_D10

RX_CD1_P R151 0R/1% MIPI_RX_CD1_P
R153 NC/0R/1% VI_D13

RX_CD1_N R155 0R/1% MIPI_RX_CD1_N
R157 NC/0R/1% VI_D12

RX_CD2_P R159 0R/1% MIPI_RX_CD2_P
R161 NC/0R/1% VI_D15

RX_CD2_N R163 0R/1% MIPI_RX_CD2_N
R165 NC/0R/1% VI_D14

RX_CD3_P R144 0R/1% MIPI_RX_CD3_P
R146 NC/0R/1% VI_D17

RX_CD3_N R148 0R/1% MIPI_RX_CD3_N
R150 NC/0R/1% VI_D16

RX_CD4_P R152 0R/1% MIPI_RX_CD4_P
R154 NC/0R/1% VI_D19

RX_CD4_N R156 0R/1% MIPI_RX_CD4_N
R158 NC/0R/1% VI_D18

RX_CD5_P R160 0R/1% MIPI_RX_CD5_P
R162 NC/0R/1% VI_D20

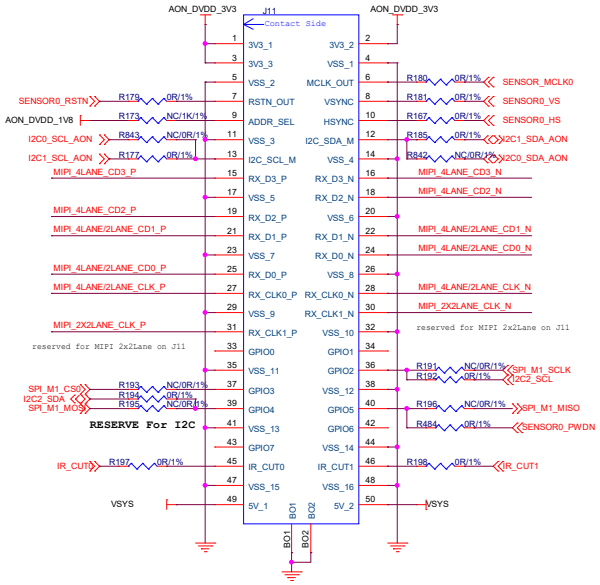
RX_CD5_N R164 0R/1% MIPI_RX_CD5_N
R166 NC/0R/1% VI_CLK1

AXERA

Page Name: MIPI_IO

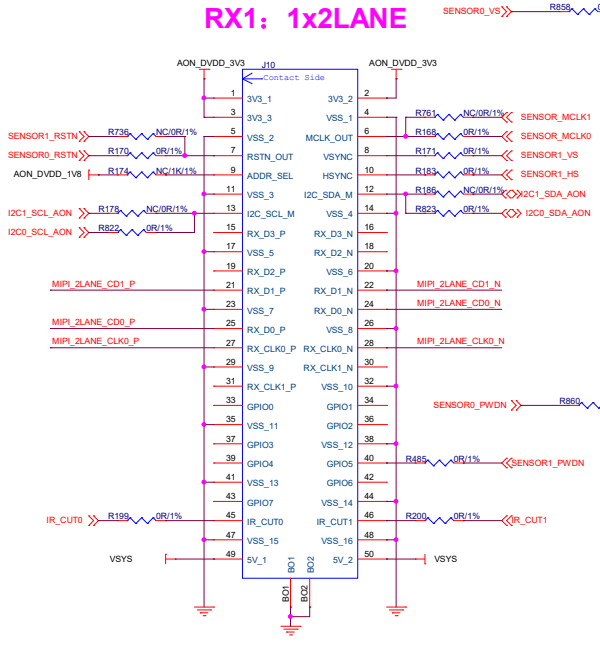
Size	Project Name - Rev	Designer	
A3	AX615D_DEMO	Checker	
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RX0：4LANE OR 2X2LANE



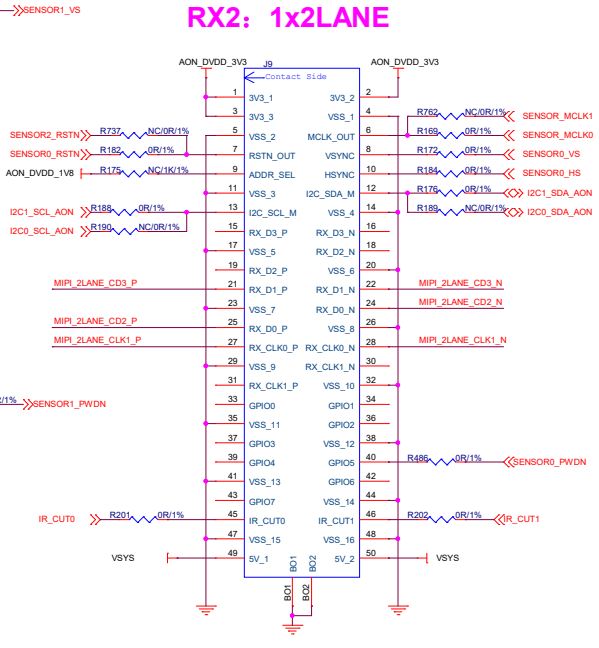
RX0：LANE0&LANE1、LANE2&LANE3

RX1：1x2LANE

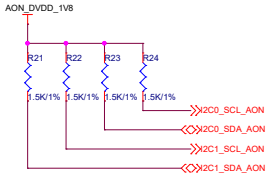


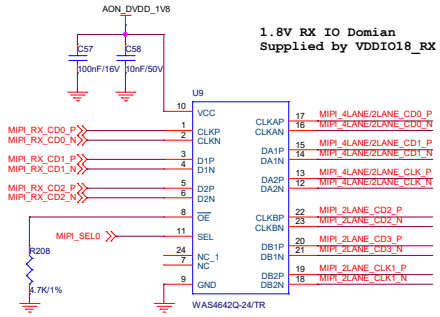
RX1：LANE2&LANE3

RX2：1x2LANE



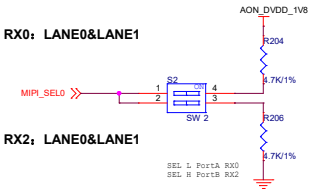
RX2：LANE0&LANE1

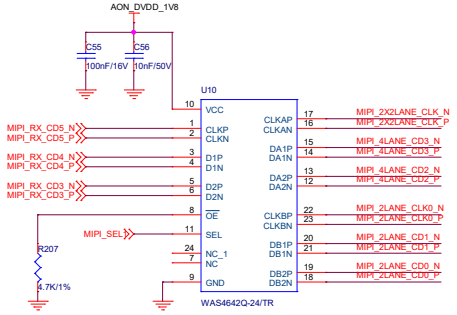




RX0：LANE0&LANE1

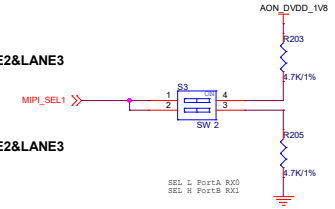
RX2：LANE0&LANE1



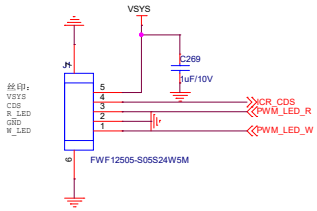


RX0：LANE2&LANE3

RX1：LANE2&LANE3



CONNECTOR TO LED-drive board



Page Name	CAM_SENSOR
Size / Project Name - Rev	A2 / AX615D_DEMO
Designer	
Checker	
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DVP 12bit

VI_D0	R209	0R/1%	DVP_DATA0
VI_D1	R210	0R/1%	DVP_DATA1
VI_D2	R212	0R/1%	DVP_DATA2
VI_D3	R214	0R/1%	DVP_DATA3
VI_D4	R215	0R/1%	DVP_DATA4
VI_D5	R217	0R/1%	DVP_DATA5
VI_D6	R219	0R/1%	DVP_DATA6
VI_D7	R221	0R/1%	DVP_DATA7
VI_D8	R230	0R/1%	DVP_HS
VI_D9	R232	0R/1%	DVP_VS
VI_D16	R234	0R/1%	DVP_DATA8
VI_D17	R228	0R/1%	DVP_DATA9
VI_D18	R237	0R/1%	DVP_DATA10
VI_D19	R239	0R/1%	DVP_DATA11
VI_CLK1	R242	0R/1%	DVP_PCLK
VI_CLK0	R243	NC/0R/1%	

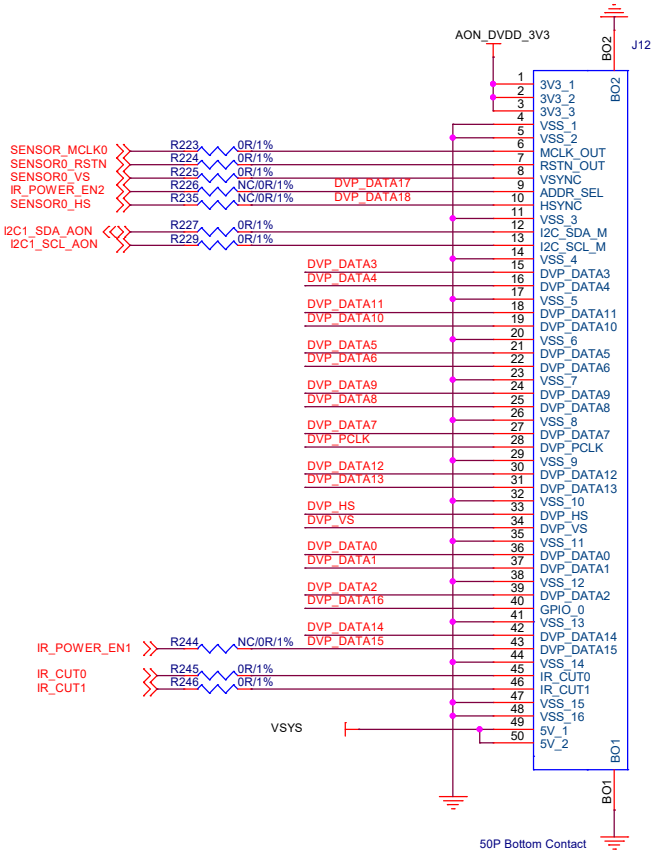
BT_IN 1120

VI_D20	R240	0R/1%	DVP_DATA12
VI_D10	R241	0R/1%	DVP_DATA13
VI_D11	R766	0R/1%	DVP_DATA14
VI_D12	R767	0R/1%	DVP_DATA15
VI_D13	R768	0R/1%	DVP_DATA16
VI_D14	R769	0R/1%	DVP_DATA17
VI_D15	R770	0R/1%	DVP_DATA18

DE

DVP 8bit

VI_D10	R211	NC/0R/1%	DVP_DATA0
VI_D11	R213	NC/0R/1%	DVP_DATA1
VI_D12	R216	NC/0R/1%	DVP_DATA2
VI_D13	R218	NC/0R/1%	DVP_DATA3
VI_D14	R220	NC/0R/1%	DVP_DATA4
VI_D15	R222	NC/0R/1%	DVP_DATA5
VI_D16	R231	NC/0R/1%	DVP_DATA6
VI_D17	R233	NC/0R/1%	DVP_DATA7
VI_D18	R236	NC/0R/1%	DVP_HS
VI_D19	R238	NC/0R/1%	DVP_VS



SENSOR_MCLK0	R223	0R/1%	
SENSOR0_RSTN	R224	0R/1%	
SENSOR0_VS	R225	0R/1%	
IR_POWER_EN2	R226	NC/0R/1%	DVP_DATA17
SENSOR0_HS	R235	NC/0R/1%	DVP_DATA18
I2C1_SDA_AON	R227	0R/1%	
I2C1_SCL_AON	R229	0R/1%	
IR_POWER_EN1	R244	NC/0R/1%	
IR_CUT0	R245	0R/1%	
IR_CUT1	R246	0R/1%	

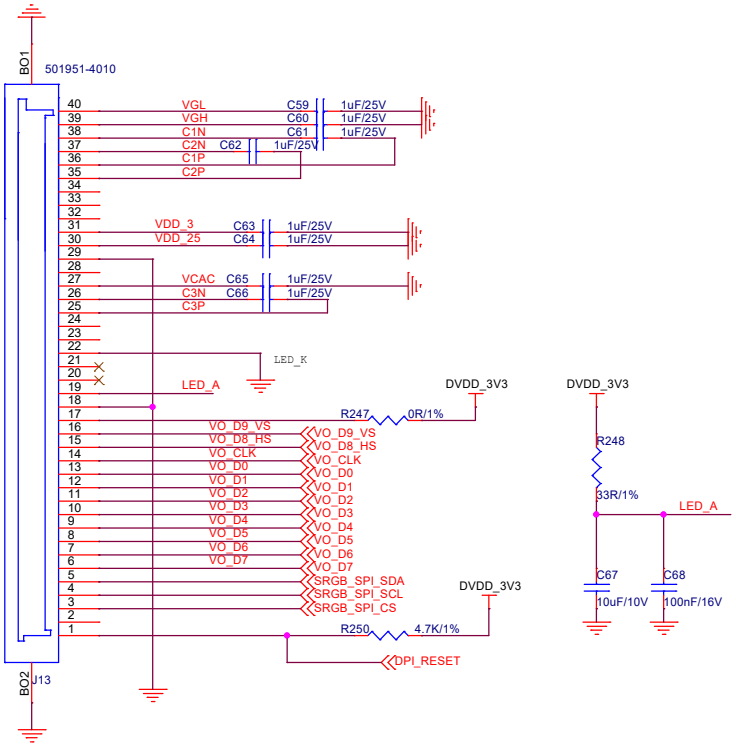
1	3V3_1	26	VSS_8
2	3V3_2	27	DVP_DATA7
3	3V3_3	28	DVP_PCLK
4	VSS_1	29	VSS_9
5	VSS_2	30	DVP_DATA12
6	MCLK_OUT	31	DVP_DATA13
7	RSTN_OUT	32	VSS_10
8	VSYNC	33	DVP_HS
9	ADDR_SEL	34	DVP_VS
10	HSYNC	35	VSS_11
11	VSS_3	36	DVP_DATA0
12	I2C_SDA_M	37	DVP_DATA1
13	I2C_SCL_M	38	VSS_12
14	VSS_4	39	DVP_DATA2
15	DVP_DATA3	40	GPIO_0
16	DVP_DATA4	41	VSS_13
17	VSS_5	42	DVP_DATA14
18	DVP_DATA11	43	DVP_DATA15
19	DVP_DATA10	44	VSS_14
20	VSS_6	45	IR_CUT0
21	DVP_DATA5	46	IR_CUT1
22	DVP_DATA6	47	VSS_15
23	VSS_7	48	VSS_16
24	DVP_DATA9	49	5V_1
25	DVP_DATA8	50	5V_2

AXERA

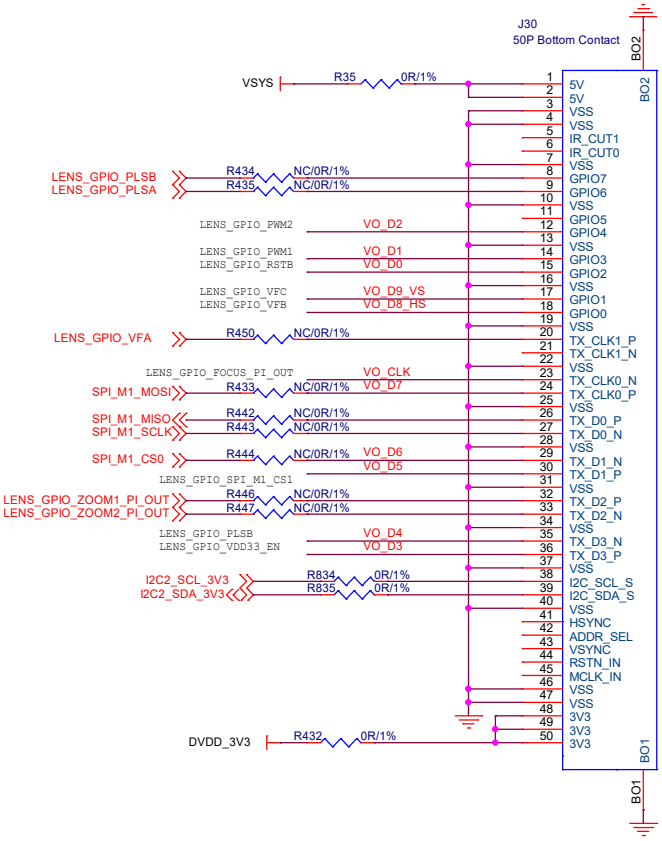
Page Name: DVP_BT_IN

Size: A3	Project Name - Rev: AX615D_DEMO	Designer: xieyibei@axera-tech.com
		Checker: hongfangjun@axera-tech.com
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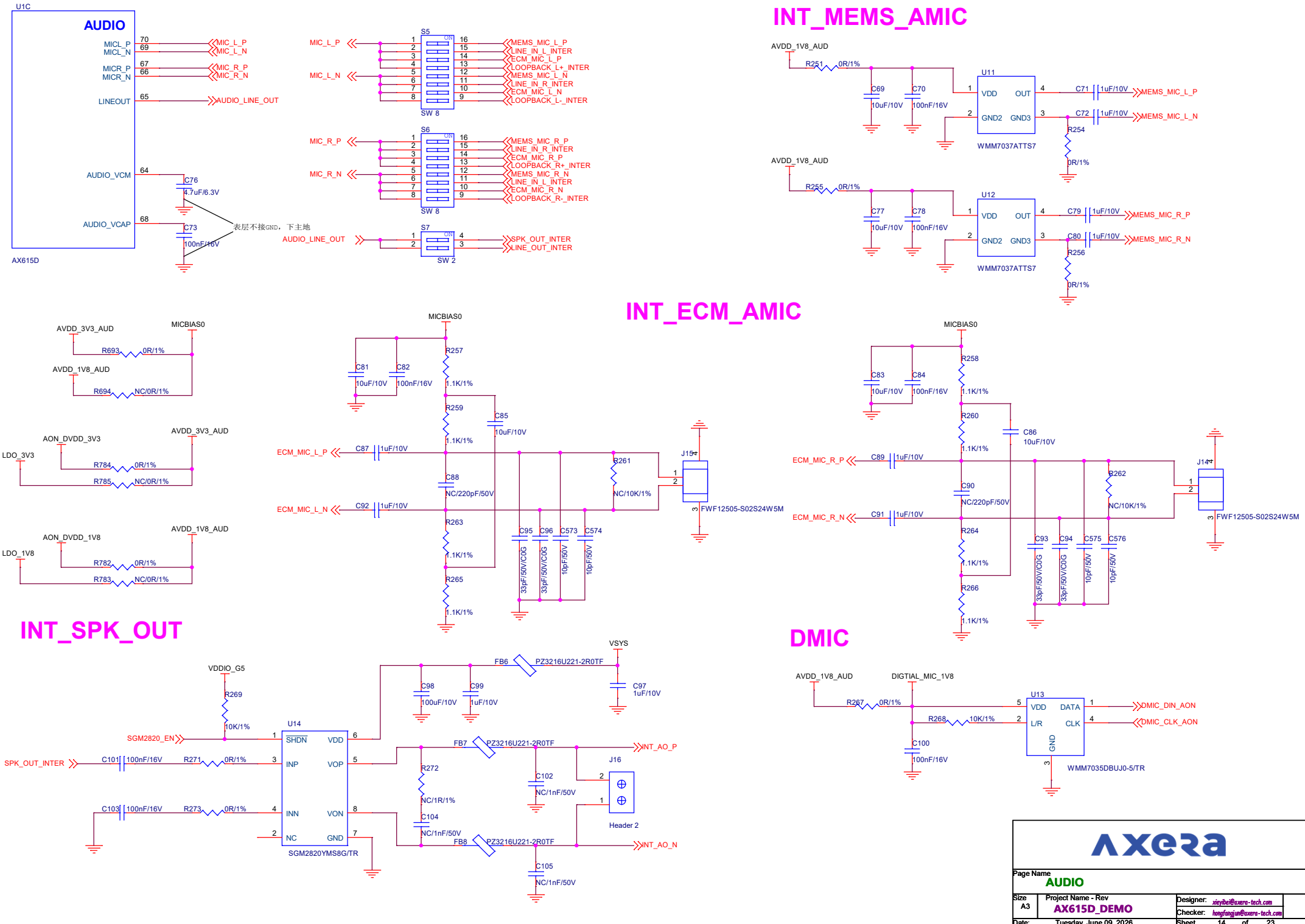
SRGB LCD



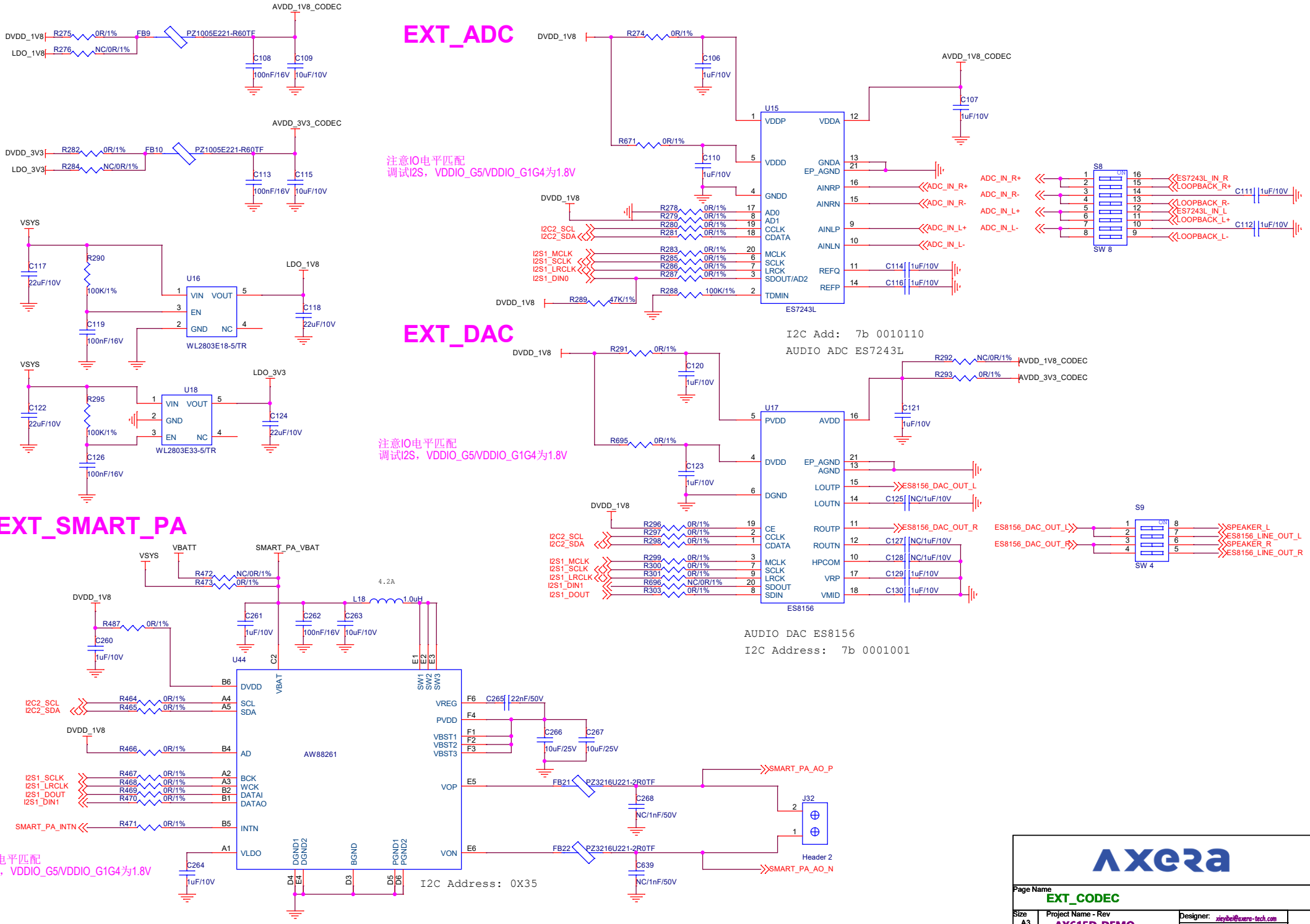
BT656 OUT & LENS DRIVER



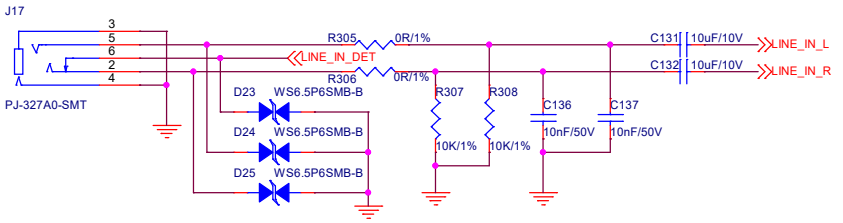
Page Name	SRGB_BT_OUT_LENS	
Size: A3	Project Name - Rev: AX615D_DEMO	Designer: xieyibei@axera-tech.com
		Checker: hongfangjun@axera-tech.com
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Page Name	AUDIO		
Size A3	Project Name - Rev AX615D_DEMO	Designer: xieyibei@axera-tech.com	
		Checker: hongfangjun@axera-tech.com	
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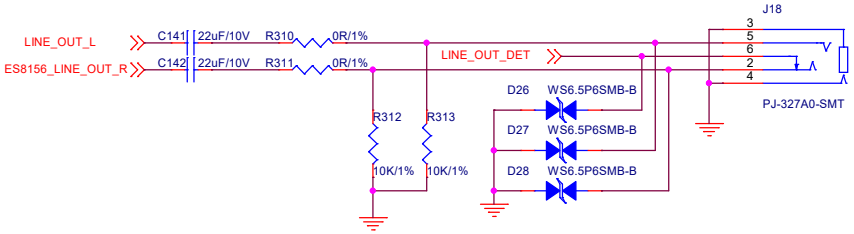

EXT_ADC
EXT_DAC
EXT_SMART_PA
AVDD_1V8_CODEC
AVDD_3V3_CODEC
DVDD_1V8
LDO_1V8
DVDD_3V3
LDO_3V3
VSYS
VBATT
SMART_PA_VBAT
U15
ES7243L
U16
WL2803E18-5/TR
U18
WL2803E33-5/TR
U17
ES8156
U44
AW88261
S8
SW 8
S9
SW 4
J32
Header 2
注意IO电平匹配
调试I2S，VDDIO_G5/VDDIO_G1G4为1.8V
I2C Add: 7b 0010110
AUDIO ADC ES7243L
AUDIO DAC ES8156
I2C Address: 7b 0001001
I2C Address: 0X35
ADC_IN_R+
ADC_IN_R-
ADC_IN_L+
ADC_IN_L-
ES8156_DAC_OUT_L
ES8156_DAC_OUT_R
SMART_PA_AO_P
SMART_PA_AO_N
SMART_PA_INTN
I2C2_SCL
I2C2_SDA
I2S1_MCLK
I2S1_SCLK
I2S1_LRCLK
I2S1_DIN0
I2S1_DIN1
I2S1_DOUT
Page Name
EXT_CODEC
Size
A3
Project Name - Rev
AX615D_DEMO
Designer:
xieyibei@axera-tech.com
Checker:
hongfangjun@axera-tech.com
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LINE_IN



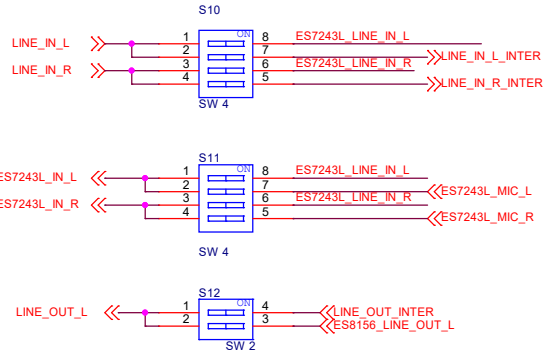
LINE_IN_DET, Internal Pull Up
LOW: not detected
HIGH: detected

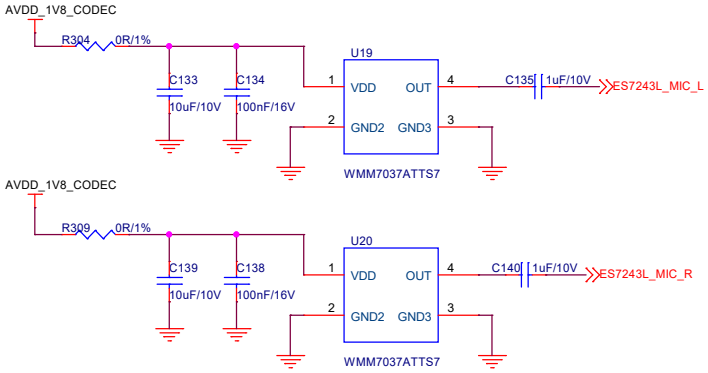
LINE_OUT



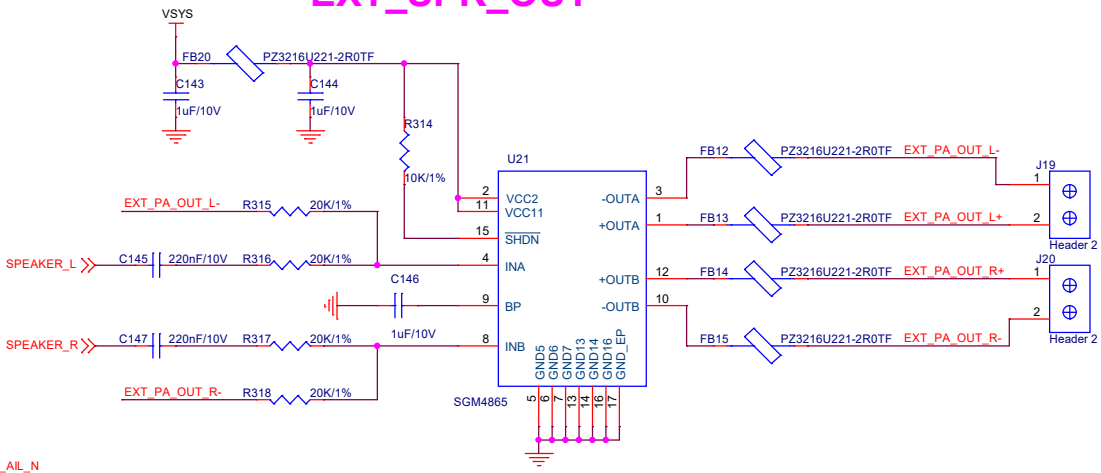
LINE_OUT_DET, Internal Pull Up
LOW: not detected
HIGH: detected

EXT_MEMS_AMIC

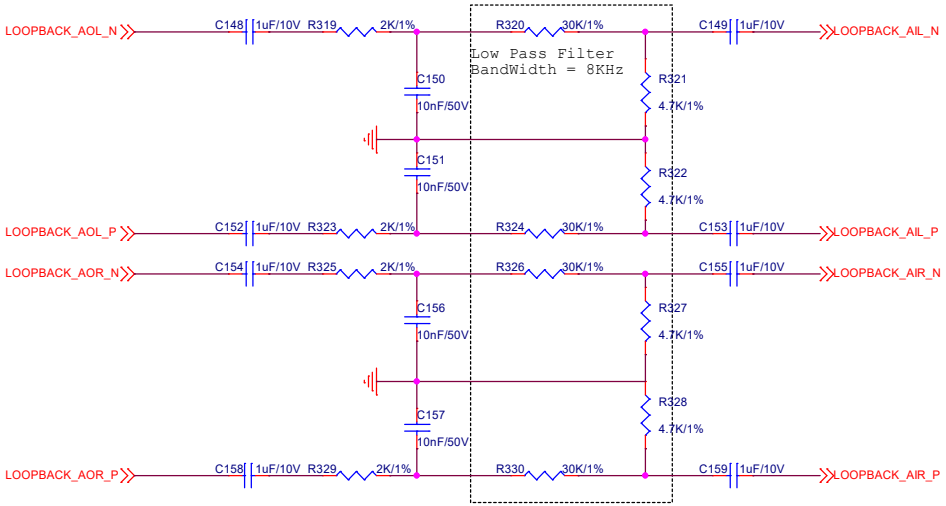


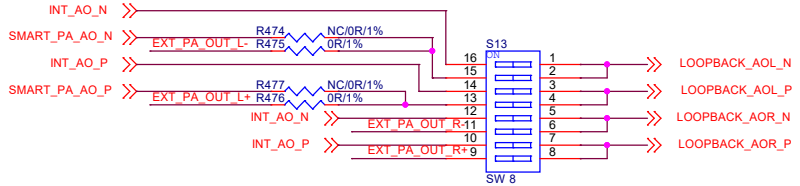


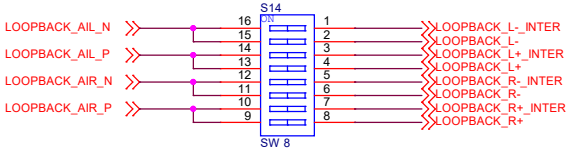
EXT_SPK_OUT



LOOPBACK





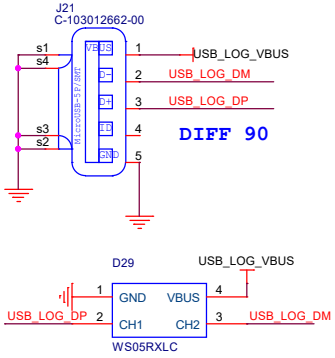


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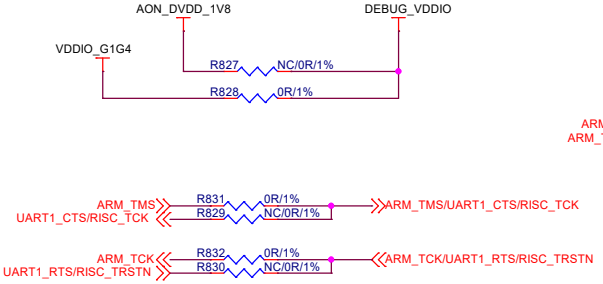
Page Name: LINE_MIC_IN_SPEAKER_OUT

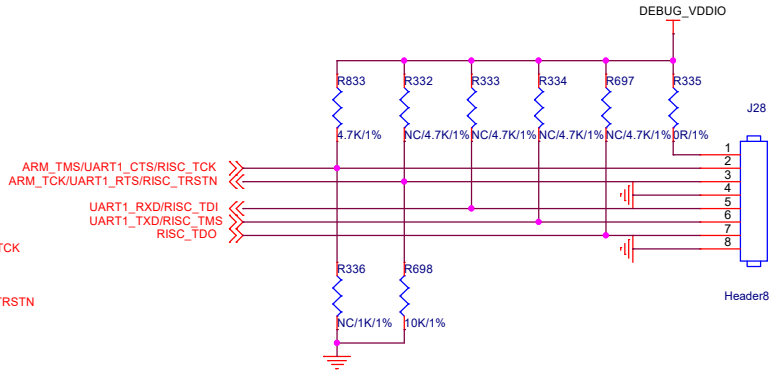
Size	Project Name - Rev	Designer: xieyibei@axera-tech.com
A3	AX615D_DEMO	Checker: hongfangjun@axera-tech.com
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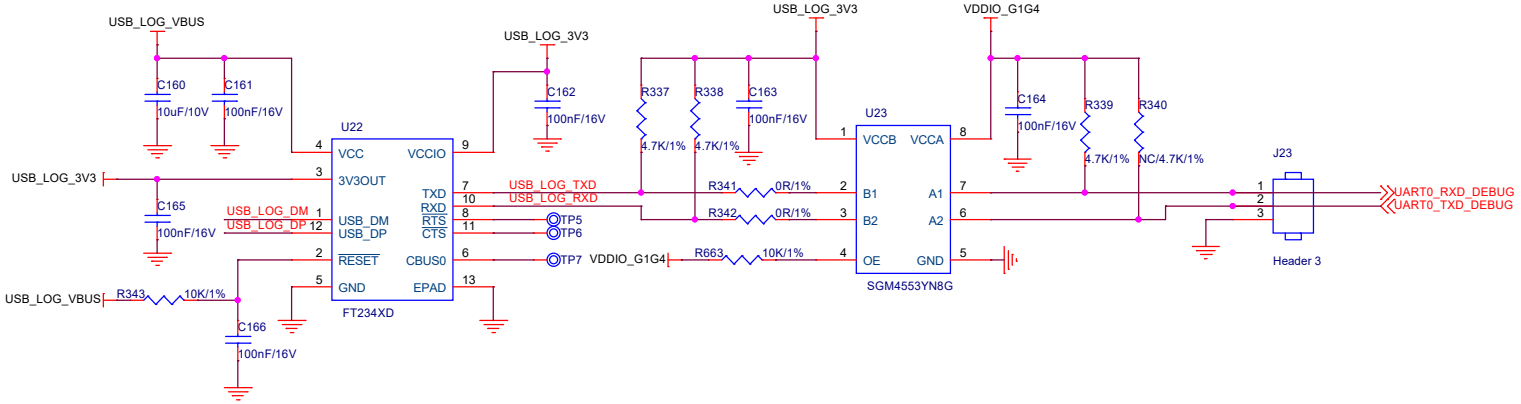
USB TO UART LOG



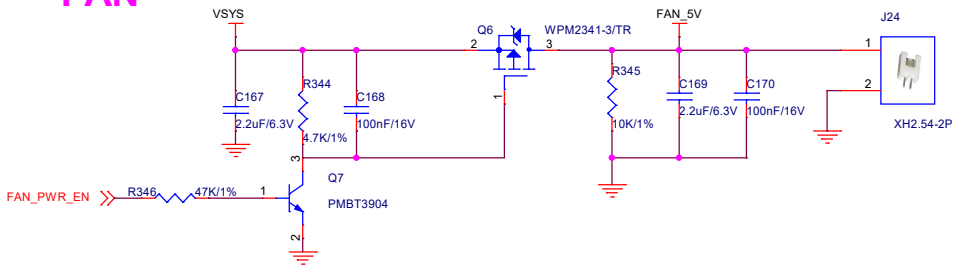
ARM_JTAG&AON UART1 & RISC JTAG







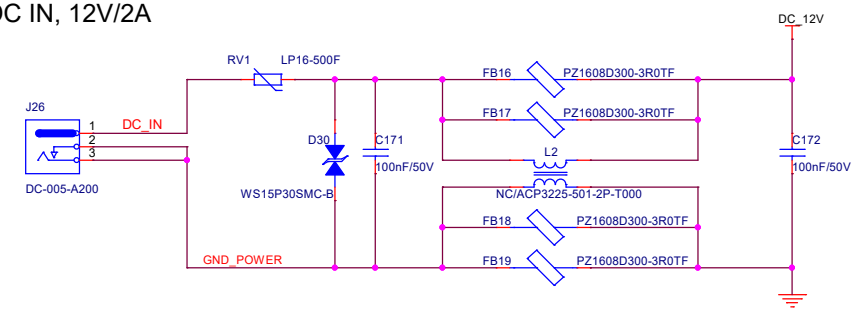
FAN

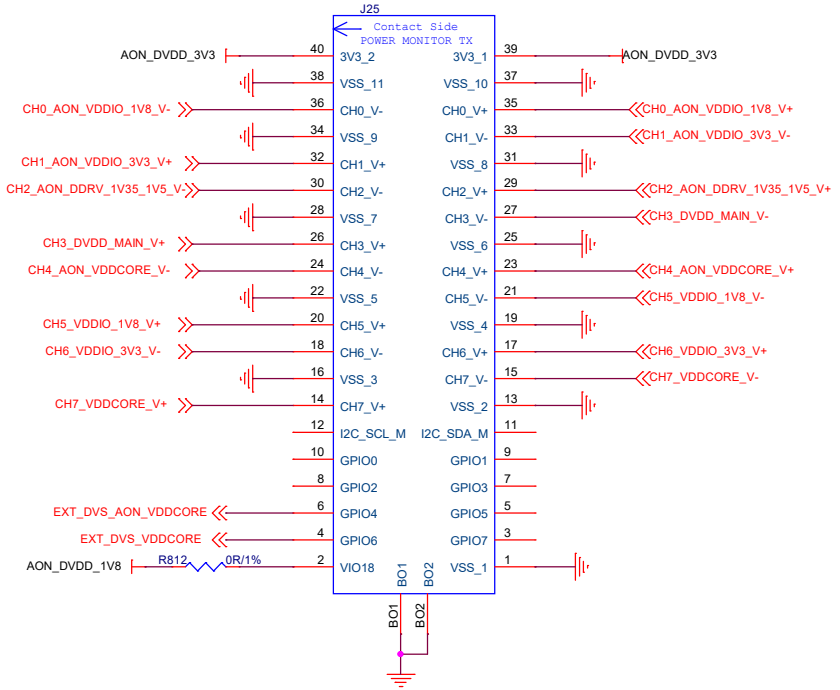


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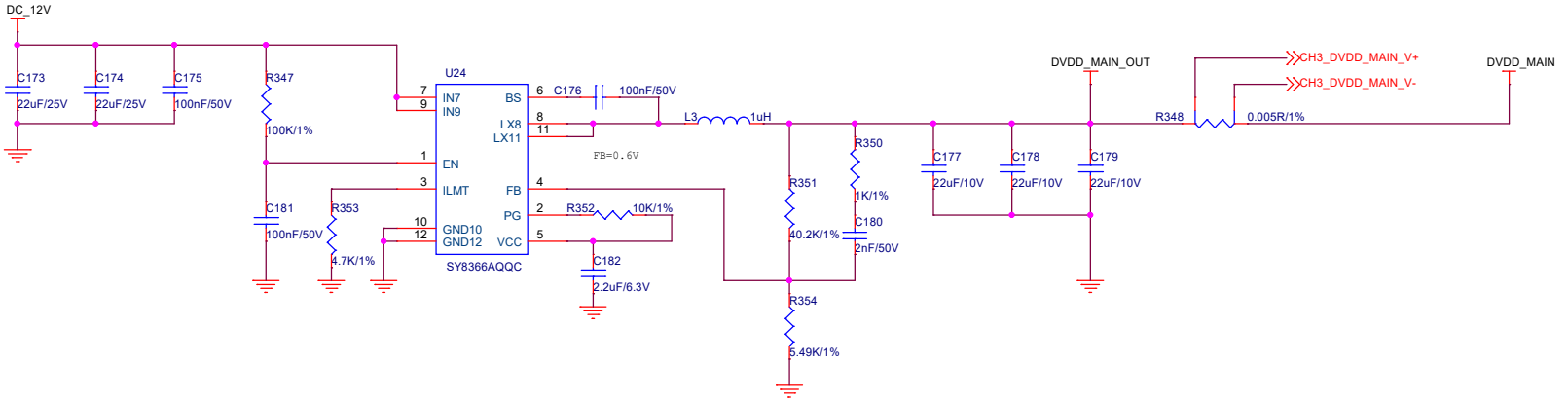
Page Name	Debug
Size	A3
Project Name - Rev	AX615D_DEMO
Designer:	xieyibei@axera-tech.com
Checker:	hongfangjun@axera-tech.com
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DC IN, 12V/2A





12V-->5V
Vout=0.6*(1+40.2K/5.49K)=4.99V



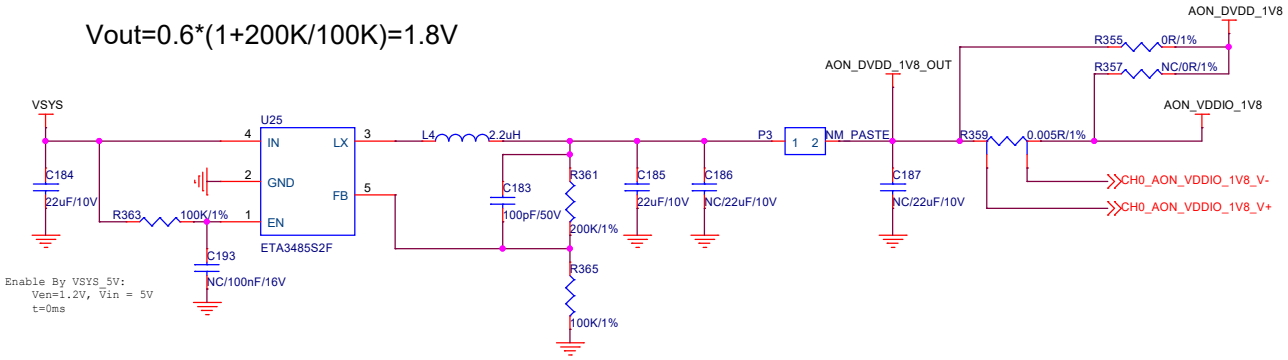
AXERA

Page Name
POWER IN

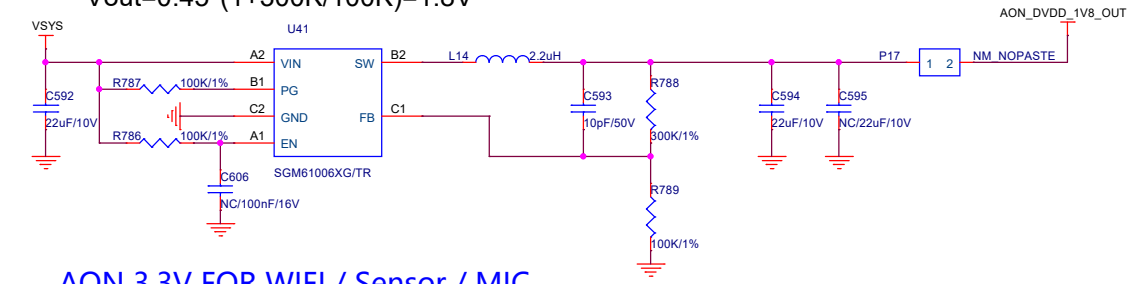
Size	Project Name - Rev	Designer: xieyibei@axera-tech.com
A3	AX615D_DEMO	Checker: hongfangjun@axera-tech.com
Date: Tuesday, June 09, 2026		Sheet 18 of 23

AON 1.8V FOR SOC / Sensor / AUDIO / (MIC)

Vout=0.6*(1+200K/100K)=1.8V

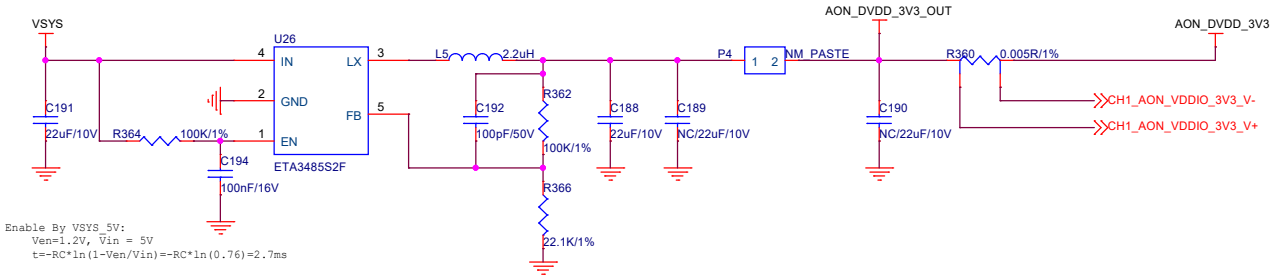


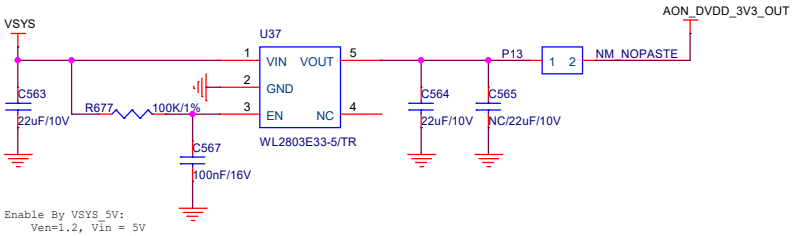
Vout=0.45*(1+300K/100K)=1.8V



AON 3.3V FOR WIFI / Sensor / MIC

Vout=0.6*(1+100K/22.1K)=3.3V



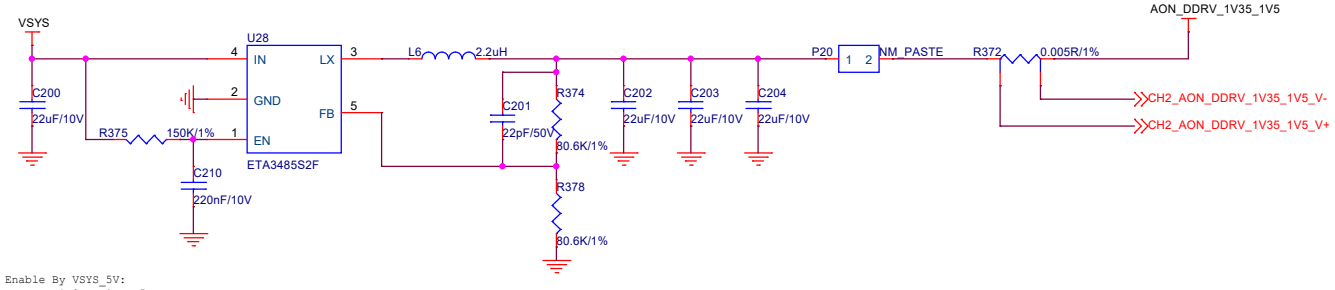


AON DDR VDDQ FOR SOC

Vout=0.6*(1+100K/80.6K)=1.35V

Vout=0.6*(1+120K/80.6K)=1.5V

Vout=0.6*(1+80.6K/80.6K)=1.2V



1.5V for DDR2L/DDR3
1.35V for DDR3L
1.2V for LPDDR3/LPDDR2

Chip PN	SIP DDR type	DRAM Size	DDR_VDDQ
AX615DV100	DDR2L	64MB	1.5V
AX615DV110	LPDDR2	64MB	1.2V
AX615DV200	DDR3L	128MB	1.35V
AX615DV201	DDR3L	256MB	1.35V
AX615DV210	LPDDR3	128MB	1.2V
AX615DV220	DDR3	128MB	1.5V

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Page Name: POWER AON IO

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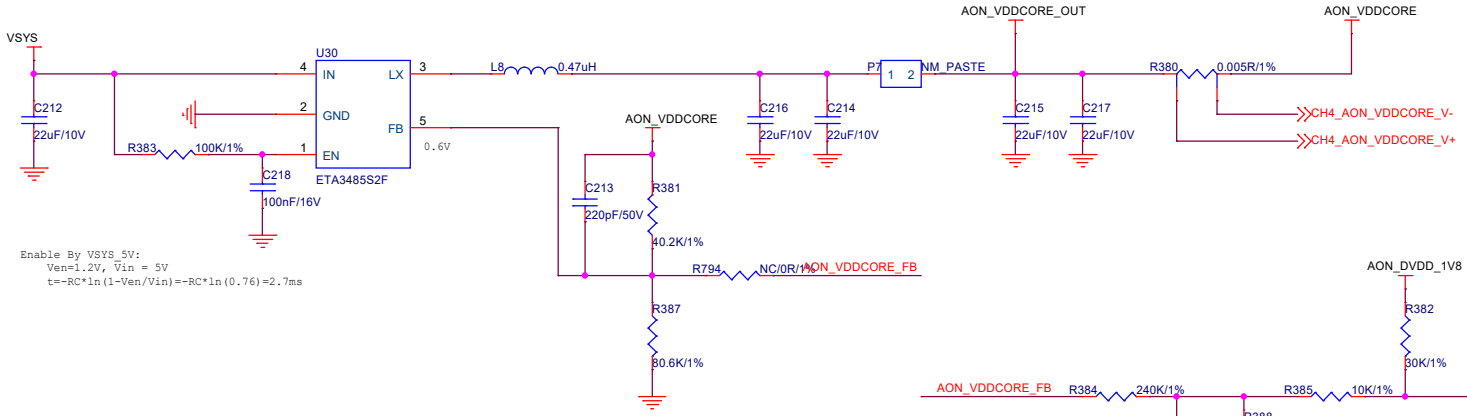
Checker: hongfangjun@axera-tech.com

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AON VDDCORE FOR SOC
VSYS-->AON_0.9V

Vout=0.6*(1+40.2K/80.6K)=0.9V

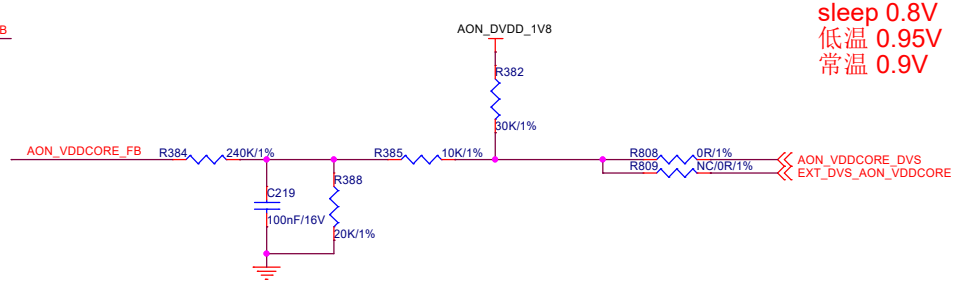


AON_VDDCORE_DVS Freq = 240KHz

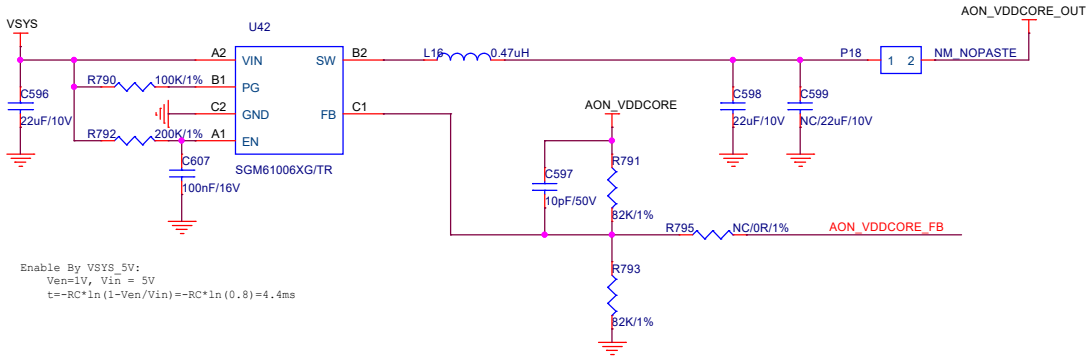
AON_VDDCORE_DVS Duty Cycle = (6-6*AON_VDDCORE)/1.2*100%

AON_VDDCORE Default = 0.9V, Tuning Range = 0.8V~1V

sleep 0.8V
低温 0.95V
常温 0.9V



Vout=0.45*(1+82K/82K)=0.9V



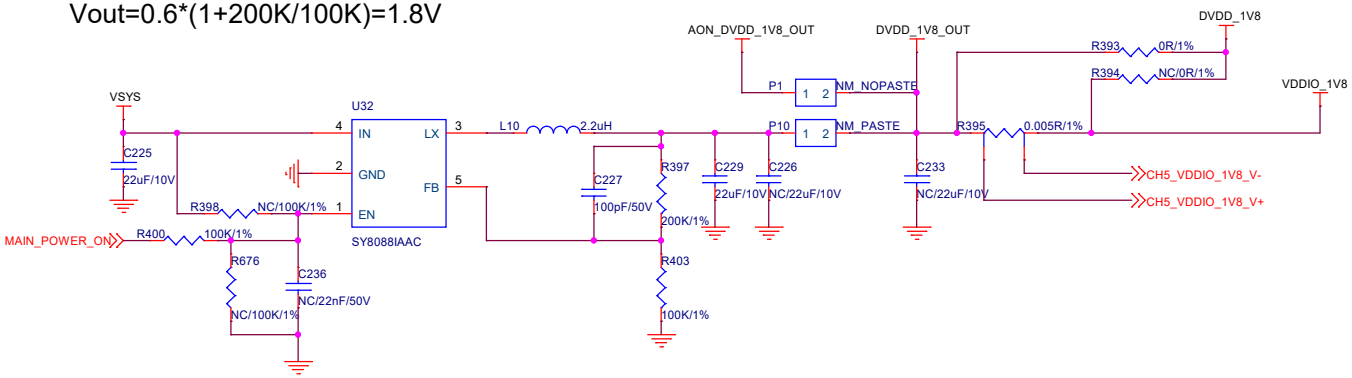
AXERA

Page Name: POWER AON CORE

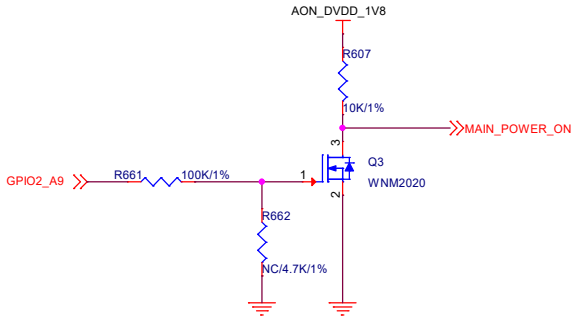
Size	Project Name - Rev	Designer: xieyibei@axera-tech.com
A3	AX615D_DEMO	Checker: hongfangjun@axera-tech.com
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MAIN 1.8V FOR SOC (ePHY)
VSYS -->1.8V

Vout=0.6*(1+200K/100K)=1.8V

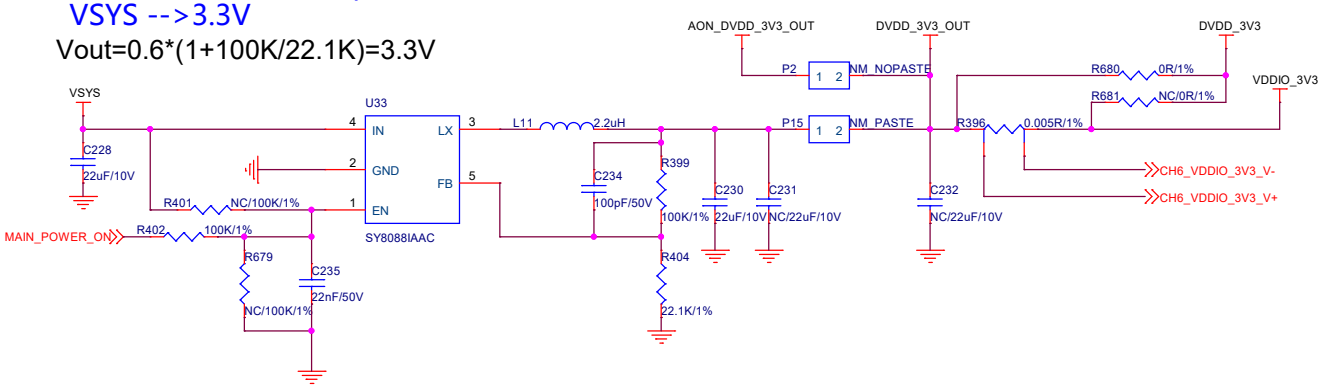


MAIN_POWER_ON out H: Enable By AON_DVDD_1V8:
Ven=1.2V, Vin = 1.8V
t=0ms



MAIN 3.3V FOR SOC / SD
VSYS -->3.3V

Vout=0.6*(1+100K/22.1K)=3.3V

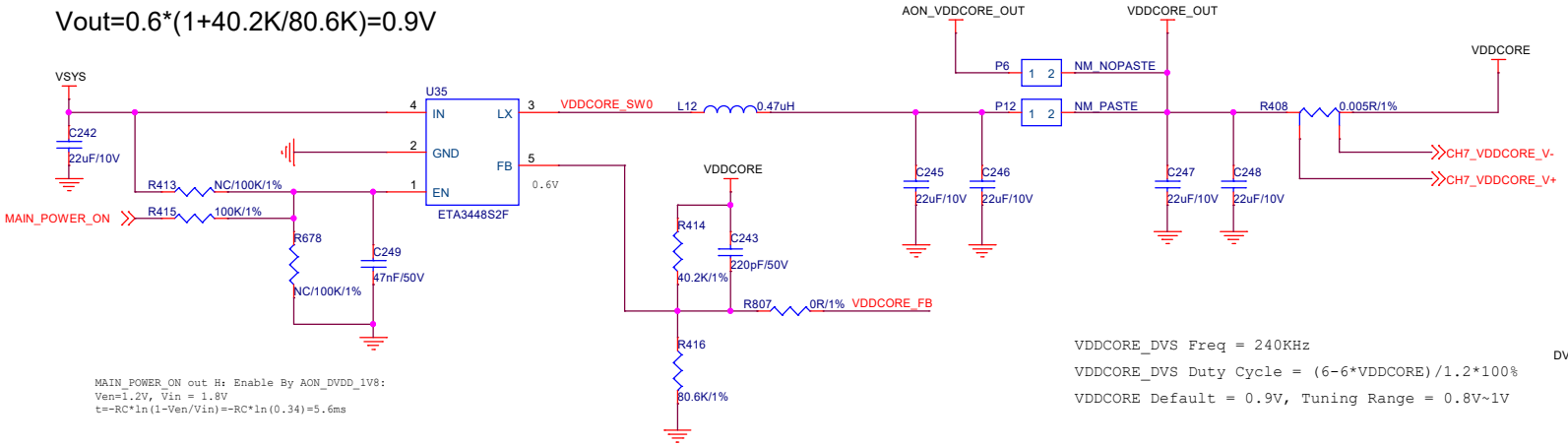


MAIN_POWER_ON out H: Enable By AON_DVDD_1V8:
Ven=1.2V, Vin = 1.8V
t=-RC*ln(1-Ven/Vin)=-RC*ln(0.34)=2.6ms

Page Name: POWER MAIN IO		
Size: A3	Project Name - Rev: AX615D_DEMO	Designer: xieyibei@axera-tech.com
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MAIN VDDCORE FOR SOC
VSYS -->0.9V

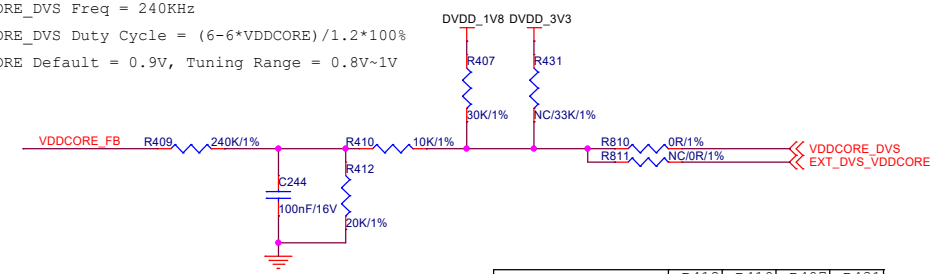
Vout=0.6*(1+40.2K/80.6K)=0.9V



MAIN_POWER_ON out H: Enable By AON_DVDD_1V8:
Ven=1.2V, Vin = 1.8V
t=-RC*ln(1-Ven/Vin)=-RC*ln(0.34)=5.6ms

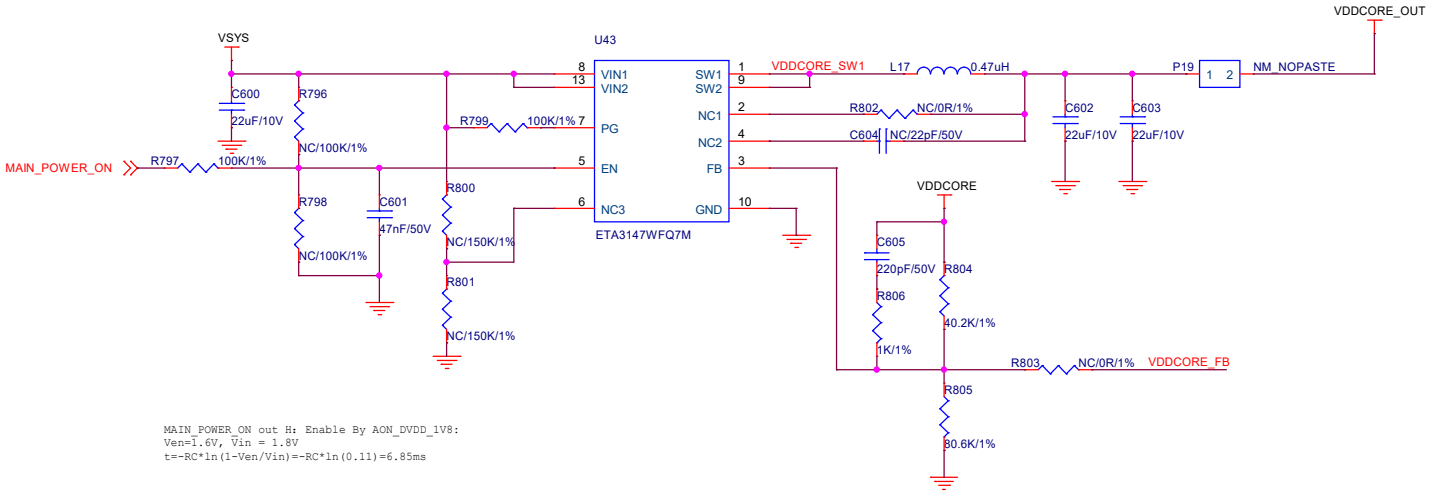
sleep 0V 低温0.95V 常温0.9V

VDDCORE_DVS Freq = 240KHz
VDDCORE_DVS Duty Cycle = (6-6*VDDCORE)/1.2*100%
VDDCORE Default = 0.9V, Tuning Range = 0.8V~1V



	R412	R410	R407	R431
VIO=1.8V(default)	20K	10K	30K	NC
VIO=3.3V	12K	21K	NC	33K

Vout=0.6*(1+40.2K/80.6K)=0.9V



MAIN_POWER_ON out H: Enable By AON_DVDD_1V8:
Ven=1.6V, Vin = 1.8V
t=-RC*ln(1-Ven/Vin)=-RC*ln(0.11)=6.85ms

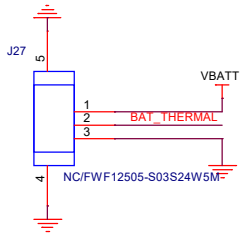
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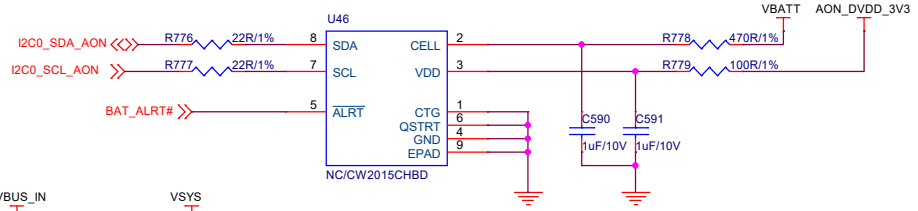
Page Name: POWER MAIN CORE

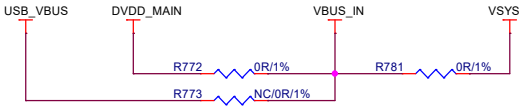
Size	Project Name - Rev	Designer	
A3	AX615D_DEMO	xieyibei@axera-tech.com	
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BAT IN
typ 3.7V

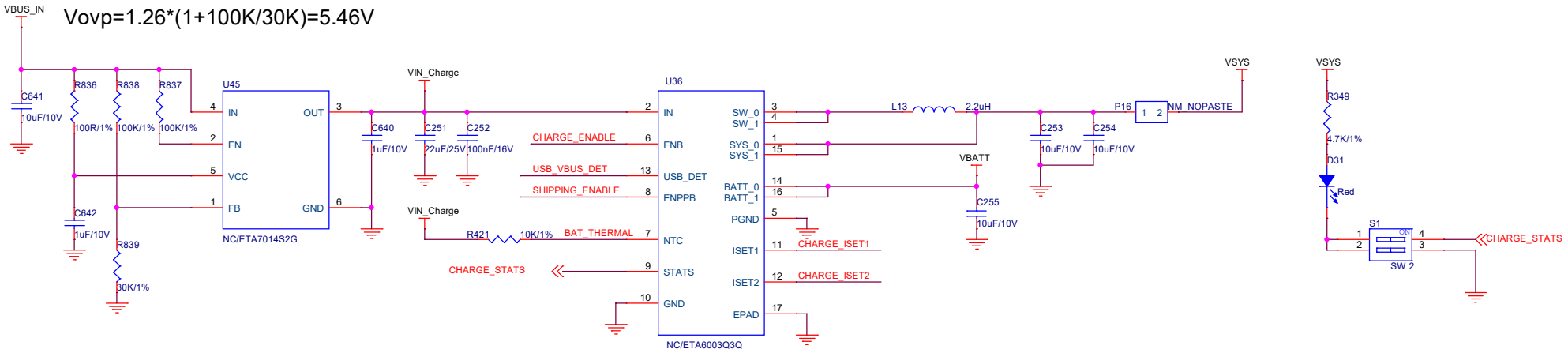
CW2015电量计电路
Gas Gauge

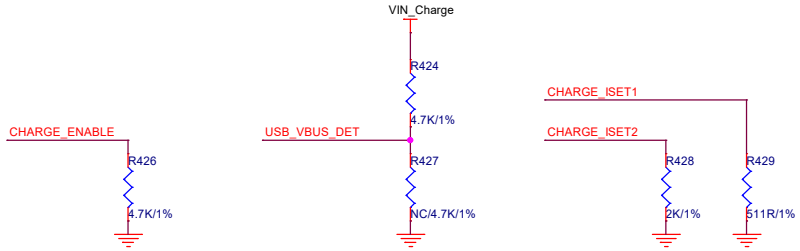


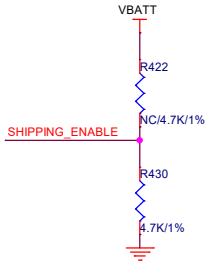


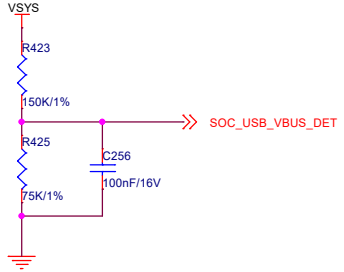


Vovp=1.26*(1+100K/30K)=5.46V









ENB = 0 Enable Charge
ENB = 1 Disable Charge

USB_DET = 0 ISET1 Set Charge Current For AC Adapter
USB_DET = 1 ISET2 Set Charge Current For USB Input

SHIPPING_ENABLE tie to VBATT Shipping MODE ON
SHIPPING_ENABLE tie to GND Shipping MODE OFF

VDET=5V*(75K+150K)/75K=1.67V

VDET=4.2V*(75K+150K)/75K=1.4V

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Page Name
BAT CHARGER

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